

CYBERSECURITY LABORATORY

Firewall • IDS/IPS • SIEM

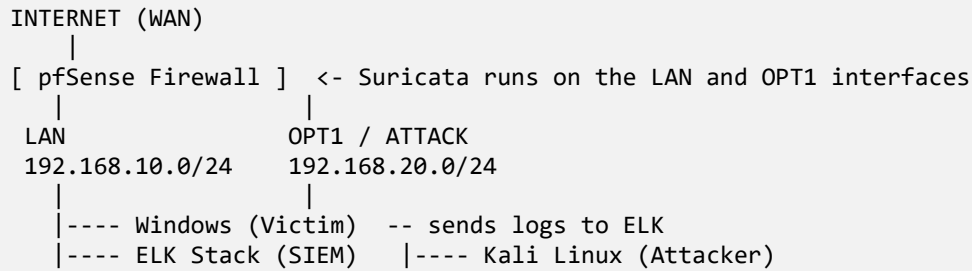
pfSense | Suricata | ELK Stack | Kali Linux | Windows

Student: Adila Yusifova

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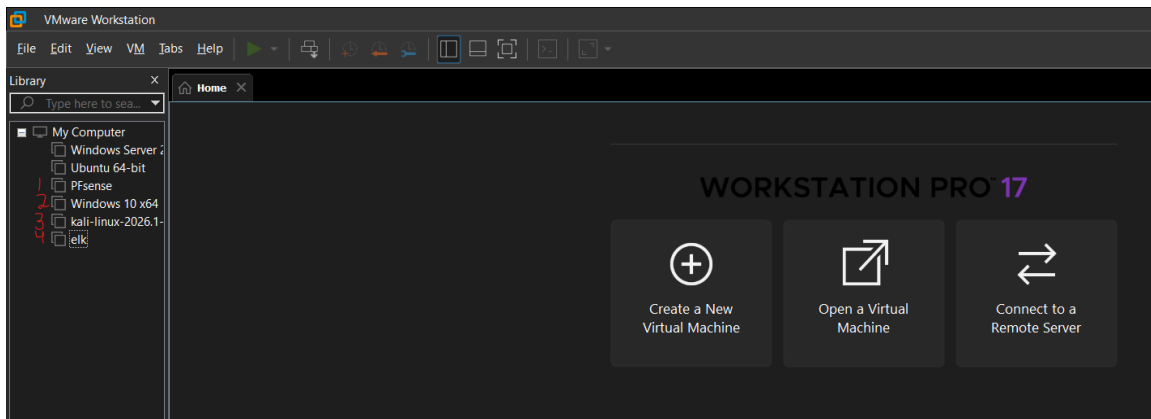
Topology



Attack path: Kali (OPT1) → pfSense (firewall + Suricata) → Windows (LAN)

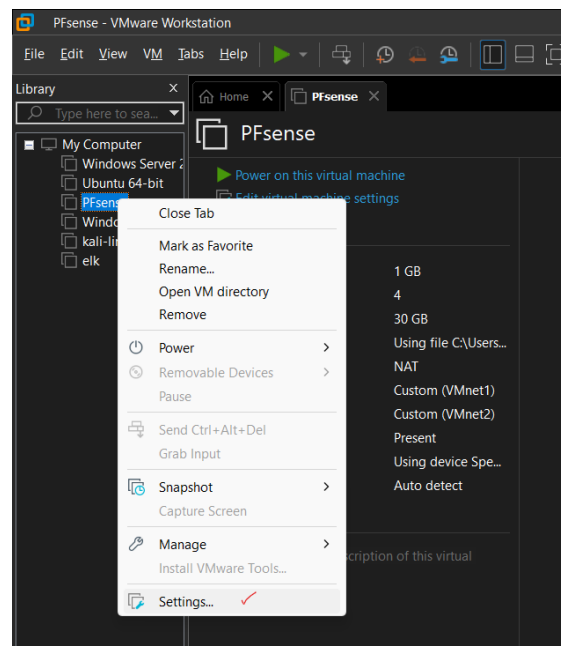
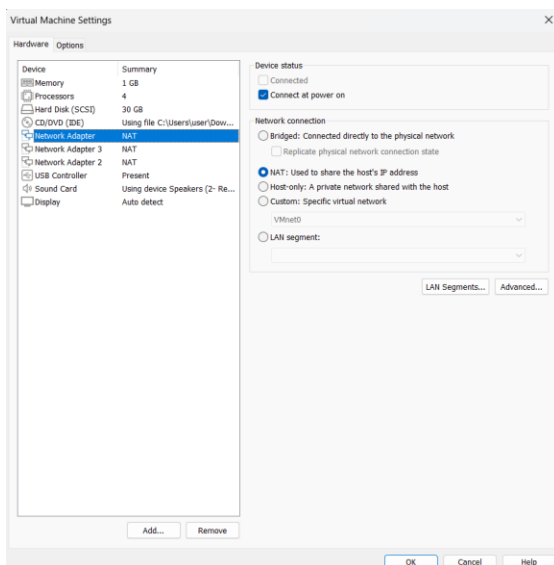
Phase 1 — Environment Setup

1. Set up a virtual environment (VirtualBox / VMware / Proxmox) and create 4 machines.



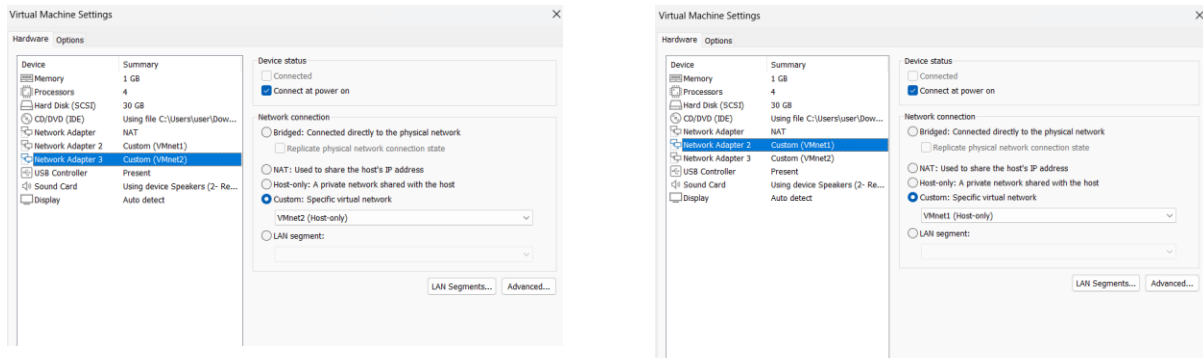
2. Install pfSense and configure 3 interfaces: WAN (internet), LAN (192.168.10.0/24), OPT1/ATTACK (192.168.20.0/24).

Right-click → Settings



Network Adapter is set to NAT mode, because pfSense's WAN interface needs to reach the internet. NAT mode gives pfSense internet access through the host computer.

Network Adapter 2 and Network Adapter 3 are set to "Custom (VMnet)" mode so that separate virtual networks are created for LAN and OPT1. With this configuration, Windows and Kali sit on different subnets, and all traffic between them passes through, and is controlled by, the pfSense firewall.



The core idea is that Kali should not be able to reach Windows directly. It must first go through pfSense, and only then reach Windows.

Network Adapter 2 → Custom: Specific virtual network → VMnet 1

Network Adapter 3 → Custom: Specific virtual network → VMnet 2

Click OK to confirm the changes.

```
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260513-1943
Bootup complete

FreeBSD/amd64 (pfSense.home.arp) (ttyv0)

VMware Virtual Machine - Netgate Device ID: dc60107a61895010dd21

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan)  -> em0 -> v4/DHCP4: 10.10.1.128/24
LAN1 (lan)  -> em1 -> v4: 10.10.2.10/24
LAN2 (opt1) -> em2 -> v4: 10.10.3.10/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: █
```

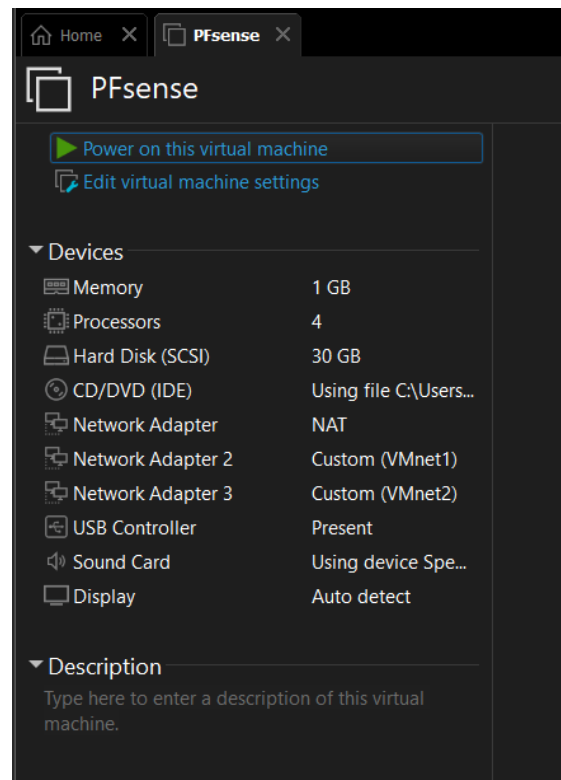
Power on this virtual machine

Enter an option: 2


```

Enter an option: 2
Available interfaces:
1 - WAN (em0 - dhcp, dhcp6)
2 - LAN1 (em1 - static)
3 - LAN2 (em2 - static)
Enter the number of the interface you wish to configure: █

```



Enter the number of the interface you wish to configure: 2 → for Windows

```

Configure IPv4 address LAN interface via DHCP? (y/n) n

```

Select "n" to enter the IPv4 address manually.

```

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.10.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) n

```

First, I enter the IPv4 address for LAN1 → 192.168.10.0 /24 → 192.168.10.1. Then I add the bit count → 24.

```

Do you want to enable the DHCP server on LAN? (y/n) y
Enter the start address of the IPv4 client address range: 192.168.10.20
Enter the end address of the IPv4 client address range: 192.168.10.200
Disabling IPv6 DHCPD...

Please wait while the changes are saved to LAN...
Reloading filter...
Reloading routing configuration...
DHCPD...

The IPv4 LAN address has been set to 192.168.10.1/24
You can now access the webConfigurator by opening the following URL in your web
browser:
    http://192.168.10.1/

Press <ENTER> to continue.

```

pfSense begins acting as a DHCP server on the LAN network; Windows will then receive an IP automatically once connected to VMnet 1.

Here I set the DHCP range → 192.168.10.20 → 192.168.10.200.

```

VMware Virtual Machine - Netgate Device ID: dc60107a61895018dd21

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan)    -> em0 -> v4/DHCP4: 10.10.1.128/24
LAN1 (lan)   -> em1 -> v4: 192.168.10.1/24
LAN2 (opt1)  -> em2 -> v4: 10.10.3.10/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option:

```

This confirms the new IP assigned for LAN1.

I repeat the same process for LAN2.

```

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN1 (em1 - static)
3 - LAN2 (em2 - static)

Enter the number of the interface you wish to configure: 3

Configure IPv4 address OPT1 interface via DHCP? (y/n) n

Enter the new OPT1 IPv4 address. Press <ENTER> for none:
> 192.168.20.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new OPT1 IPv4 subnet bit count (1 to 32):
> 24

```

First, I enter the IPv4 address for LAN2 → 192.168.20.0 /24 → 192.168.20.1. Then I add the bit count → 24.

```

Enter the new OPT1 IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new OPT1 IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address OPT1 interface via DHCP6? (y/n) n

Enter the new OPT1 IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on OPT1? (y/n) y
Enter the start address of the IPv4 client address range: 192.168.20.20
Enter the end address of the IPv4 client address range: 192.168.20.200

```

```

VMware Virtual Machine - Netgate Device ID: dc60107a61095018dd21

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 10.10.1.128/24
LAN1 (lan) -> em1 -> v4: 192.168.10.1/24
LAN2 (opt1) -> em2 -> v4: 192.168.20.1/24

0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell

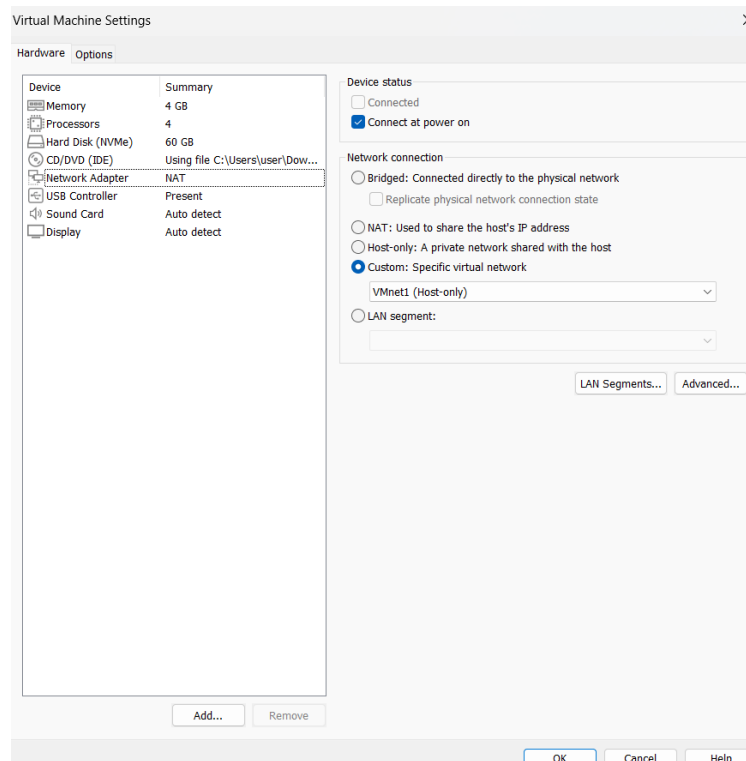
9) pfTop
10) Filter Logs
11) Restart GUI
12) PHP shell + pfSense tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM

Enter an option:

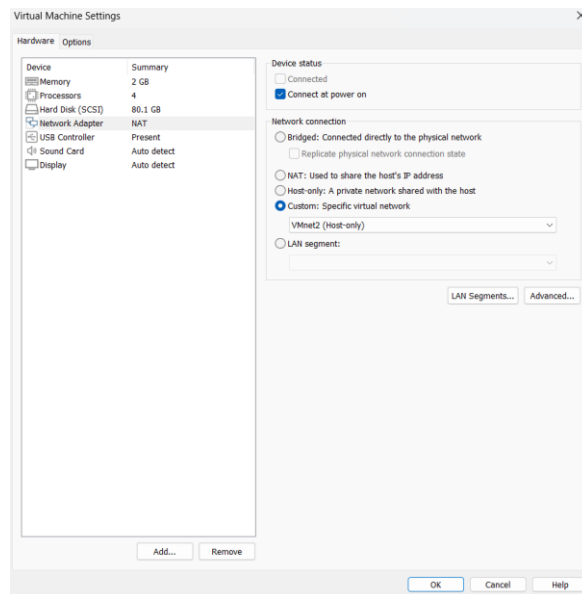
```

Here I set the DHCP range → 192.168.20.20 → 192.168.20.200. This confirms the new IP assigned for LAN2.

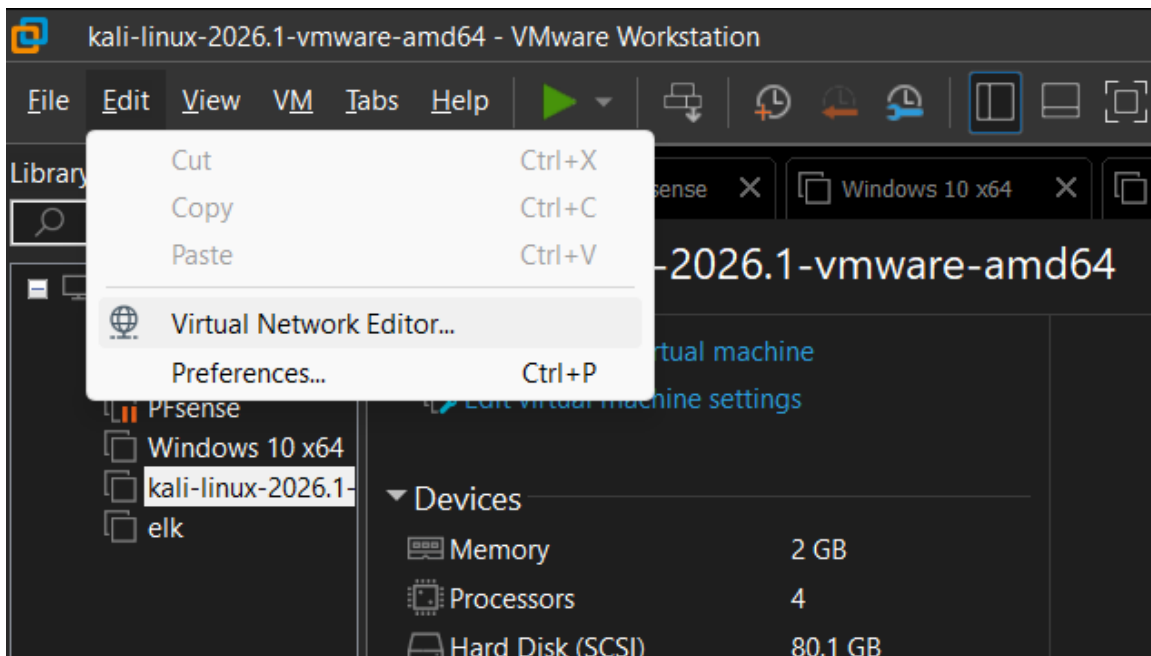
3. Connect Windows and ELK to the LAN, and Kali to the OPT1 network. (Kali and Windows must be on different subnets!)



Windows → right-click → Settings. I connect the Network Adapter to VMnet1 — pfSense reaches Windows through VMnet1. I also connect ELK to VMnet1, and Kali Linux to VMnet2.



4. Enable Promiscuous / "Allow All" mode on the VM adapters and set the adapter type to E1000 (required for IPS mode).



Edit → Virtual Network Editor → Change Settings (requires UAC)

Virtual Network Editor

✕

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Custom	-	-	-	192.168.72.0
VMnet1	Host-only	-	Connected	-	10.10.2.0
VMnet2	Host-only	-	Connected	-	10.10.3.0
VMnet8	NAT	NAT	Connected	Enabled	10.10.1.0

Add Network...

Remove Network

Rename Network...

VMnet Information

☐ Bridged (connect VMs directly to the external network)

Bridged to:

Automatic Settings...

☐ NAT (shared host's IP address with VMs)

NAT Settings...

☒ Host-only (connect VMs internally in a private network)

☐ Connect a host virtual adapter to this network

Host virtual adapter name: VMware Network Adapter VMnet0

☐ Use local DHCP service to distribute IP address to VMs

DHCP Settings...

Subnet IP:

192 . 168 . 72 . 0

Subnet mask:

255 . 255 . 255 . 0

⚠

 Administrator privileges are required to modify the network configuration.

Change Settings

Restore Defaults

Import...

Export...

OK

Cancel

Apply

Help

Virtual Network Editor

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Custom	-	-	-	192.168.72.0
VMnet1	Host-only	-	Connected	-	192.168.10.0
VMnet2	Host-only	-	Connected	-	192.168.20.0
VMnet8	NAT	NAT	Connected	Enabled	10.10.1.0

Add Network... Remove Network Rename Network...

VMnet Information

☐ Bridged (connect VMs directly to the external network)
 Bridged to: Automatic Settings...

☐ NAT (shared host's IP address with VMs) NAT Settings...

☒ Host-only (connect VMs internally in a private network)

☒ Connect a host virtual adapter to this network
 Host virtual adapter name: VMware Network Adapter VMnet2

☐ Use local DHCP service to distribute IP address to VMs DHCP Settings...

Subnet IP: Subnet mask:

Restore Defaults Import... Export... OK Cancel Apply Help

I changed the subnet IP and subnet mask.

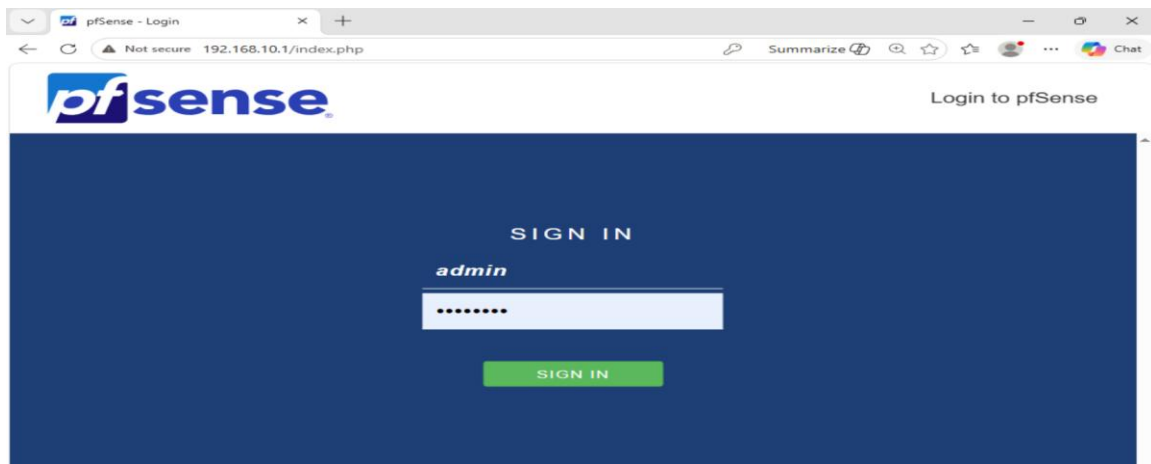
Apply → OK

```

extendedConfigFile = "PfSense.vmx"
sound.fileName = "Speakers (2- Realtek(R) Audio)"
numvcpus = "4"
ethernet1.connectionType = "custom"
ethernet1.addressType = "generated"
ethernet1.virtualDev = "e1000"
ethernet2.connectionType = "custom"
ethernet2.addressType = "generated"
ethernet2.virtualDev = "e1000"
ethernet3.connectionType = "nat"
ethernet3.addressType = "generated"
ethernet3.virtualDev = "e1000"
ethernet0.address = "00:50:56:2C:70:01"
gui.stretchGuestMode = "fullfill"
vmxstats.filename = "PfSense.scoreboard"
numa.autosize.cookie = "40022"
numa.autosize.vcpu.maxPerVirtualNode = "4"

```

E1000 is the Ethernet card model. Opening the pfSense .vmx file confirms the adapter is set to E1000.



5. Verify that all machines can "ping" each other, and that Windows only reaches the internet through pfSense.

Windows → 192.168.10.20 → Gateway: 192.168.10.1

Kali Linux → 192.168.20.20 → Gateway: 192.168.20.1

ELK → 192.168.10.22 → Gateway: 192.168.10.1

```
PS C:\Users\A> ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Request timed out.
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\A>
```

Windows → pfSense: OK

```
PS C:\Users\A> ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=49ms TTL=127
Reply from 8.8.8.8: bytes=32 time=56ms TTL=127
Reply from 8.8.8.8: bytes=32 time=38ms TTL=127
Reply from 8.8.8.8: bytes=32 time=66ms TTL=127

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 38ms, Maximum = 66ms, Average = 52ms
```

pfSense → Internet: OK

```
PS C:\Users\A> tracert 8.8.8.8

Tracing route to 8.8.8.8 over a maximum of 30 hops

  1  <1 ms    <1 ms    <1 ms    192.168.10.1
  2  1 ms     <1 ms    1 ms     10.10.1.2
  3  *         *         *         Request timed out.
  4  *         *         *         Request timed out.
  5  *         *         *         Request timed out.
  6  *         *         *         Request timed out.
  7  *         *         *         Request timed out.
  8  *         *         *         Request timed out.
  9  *         *         *         Request timed out.
 10 46 ms     37 ms    38 ms     8.8.8.8

Trace complete.
```

Windows (192.168.10.20) reaches the internet through pfSense (192.168.10.1). OK

```

PS C:\Users\A> ping 192.168.10.22

Pinging 192.168.10.22 with 32 bytes of data:
Reply from 192.168.10.22: bytes=32 time<1ms TTL=64
Reply from 192.168.10.22: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.10.22:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
PS C:\Users\A>

```

Windows → ELK: OK

```

(kali㉿kali)-[~/Desktop]
$ ping 192.168.20.1
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
64 bytes from 192.168.20.1: icmp_seq=1 ttl=64 time=0.737 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=64 time=1.02 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=64 time=0.999 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=64 time=0.875 ms
64 bytes from 192.168.20.1: icmp_seq=5 ttl=64 time=0.969 ms
64 bytes from 192.168.20.1: icmp_seq=6 ttl=64 time=0.692 ms
^C
— 192.168.20.1 ping statistics —
6 packets transmitted, 6 received, 0% packet loss, time 5013ms
rtt min/avg/max/mdev = 0.692/0.881/1.017/0.126 ms

```

Kali Linux → pfSense: OK

```

(kali㉿kali)-[~/Desktop]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=127 time=110 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=127 time=93.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=127 time=94.7 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=127 time=94.4 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=127 time=92.0 ms
^C
— 8.8.8.8 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 3996ms
rtt min/avg/max/mdev = 92.046/96.982/110.127/6.634 ms

```

pfSense → Internet: OK

```

(kali㉿kali)-[~/Desktop]
$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  192.168.20.1 (192.168.20.1)  2.457 ms  2.550 ms  3.092 ms
 2  10.10.1.2 (10.10.1.2)  8.260 ms  8.203 ms  8.164 ms
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *

```

Kali Linux (192.168.20.20) reaches the internet through pfSense (192.168.20.1). OK


```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Allow ICMPv4 Ping" -Protocol ICMPv4 -IcmpType 8 -Action Allow

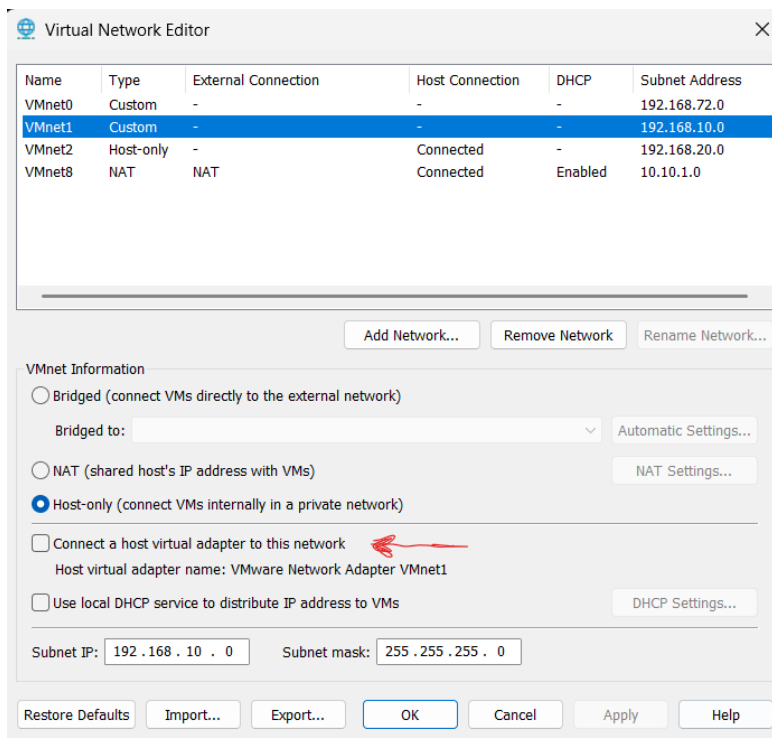
Name : {ab1dc963-0a58-49f4-9c4d-78c455a2437f}
DisplayName : Allow ICMPv4 Ping
Description :
DisplayGroup :
Group :
Enabled : True
Profile : Any
Platform : {}
Direction : Inbound
Action : Allow
EdgeTraversalPolicy : Block
LooseSourceMapping : False
LocalOnlyMapping : False
Owner :
PrimaryStatus : OK
Status : The rule was parsed successfully from the store. (65536)
EnforcementStatus : NotApplicable
PolicyStoreSource : PersistentStore
PolicyStoreSourceType : Local
RemoteDynamicKeywordAddresses :
PolicyAppId :
```

I opened Windows PowerShell as Administrator.

```
elk@elk:~$ ping 192.168.10.20
PING 192.168.10.20 (192.168.10.20) 56(84) bytes of data.
64 bytes from 192.168.10.20: icmp_seq=1 ttl=128 time=0.805 ms
64 bytes from 192.168.10.20: icmp_seq=2 ttl=128 time=0.699 ms
64 bytes from 192.168.10.20: icmp_seq=3 ttl=128 time=0.803 ms
64 bytes from 192.168.10.20: icmp_seq=4 ttl=128 time=0.642 ms
64 bytes from 192.168.10.20: icmp_seq=5 ttl=128 time=0.621 ms
64 bytes from 192.168.10.20: icmp_seq=6 ttl=128 time=0.636 ms
64 bytes from 192.168.10.20: icmp_seq=7 ttl=128 time=1.56 ms
64 bytes from 192.168.10.20: icmp_seq=8 ttl=128 time=0.723 ms
64 bytes from 192.168.10.20: icmp_seq=9 ttl=128 time=0.636 ms
^C
```

ELK → Windows: OK

However, ping from ELK to pfSense fails. EDIT → VIRTUAL NETWORK EDITOR



I unchecked "Connect a host virtual adapter to this network."

VMnet1 is for VMs only — the host machine is not part of it; only pfSense holds 192.168.10.1. VMnet2 is for VMs only — the host machine is not part of it; only pfSense holds 192.168.20.1.

```

PS C:\Users\A> ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:
Reply from 192.168.20.20: bytes=32 time=1ms TTL=63
Reply from 192.168.20.20: bytes=32 time=1ms TTL=63
Reply from 192.168.20.20: bytes=32 time=2ms TTL=63
Reply from 192.168.20.20: bytes=32 time=2ms TTL=63

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
PS C:\Users\A>

```

Windows → Kali Linux: OK

```

(kali㉿kali)-[~/Desktop]
$ ping 192.168.10.23
PING 192.168.10.23 (192.168.10.23) 56(84) bytes of data.
64 bytes from 192.168.10.23: icmp_seq=1 ttl=127 time=1.92 ms
64 bytes from 192.168.10.23: icmp_seq=2 ttl=127 time=1.23 ms
64 bytes from 192.168.10.23: icmp_seq=3 ttl=127 time=2.14 ms
64 bytes from 192.168.10.23: icmp_seq=4 ttl=127 time=1.25 ms
64 bytes from 192.168.10.23: icmp_seq=5 ttl=127 time=1.66 ms
64 bytes from 192.168.10.23: icmp_seq=6 ttl=127 time=1.45 ms
64 bytes from 192.168.10.23: icmp_seq=7 ttl=127 time=1.27 ms
^C
— 192.168.10.23 ping statistics —
7 packets transmitted, 7 received, 0% packet loss, time 5994ms
rtt min/avg/max/mdev = 1.233/1.559/2.136/0.331 ms

(kali㉿kali)-[~/Desktop]
$

```

Kali Linux → Windows: OK (the IP changed after renewing the MAC address)

```

(kali㉿kali)-[~/Desktop]
$ ping 192.168.10.22
PING 192.168.10.22 (192.168.10.22) 56(84) bytes of data.
64 bytes from 192.168.10.22: icmp_seq=1 ttl=63 time=2.93 ms
64 bytes from 192.168.10.22: icmp_seq=2 ttl=63 time=1.05 ms
64 bytes from 192.168.10.22: icmp_seq=3 ttl=63 time=1.46 ms
64 bytes from 192.168.10.22: icmp_seq=4 ttl=63 time=1.09 ms
64 bytes from 192.168.10.22: icmp_seq=5 ttl=63 time=1.03 ms
^C
— 192.168.10.22 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 3996ms
rtt min/avg/max/mdev = 1.030/1.511/2.925/0.723 ms

```

Kali Linux → ELK: OK

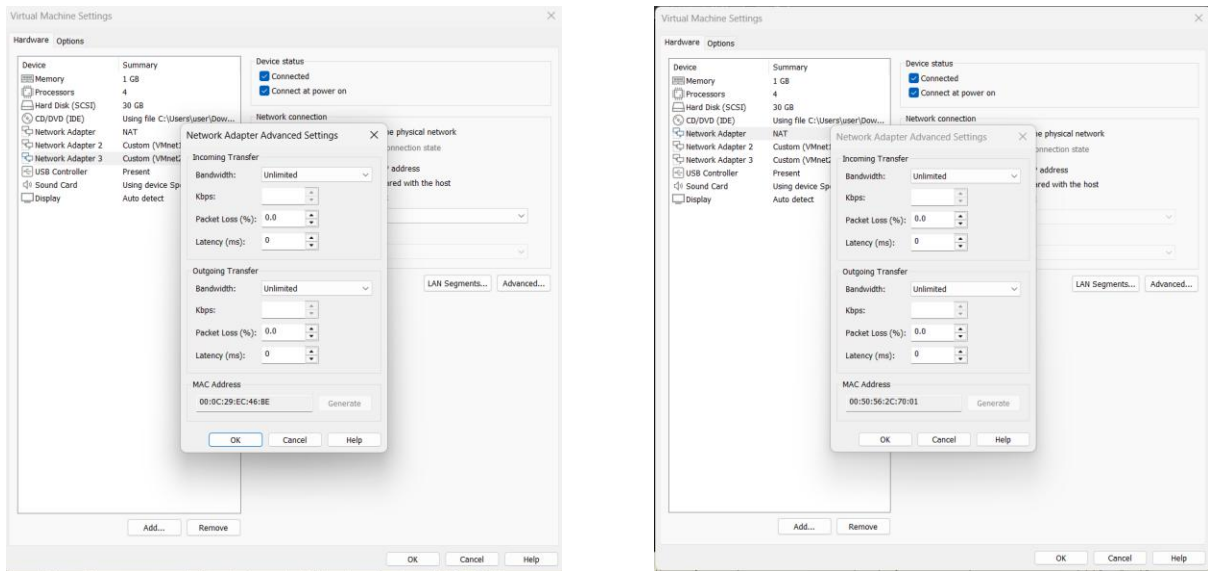
Phase 2 — Firewall Configuration (pfSense)

6. Allow internet access from LAN to WAN (the default rule is sufficient).

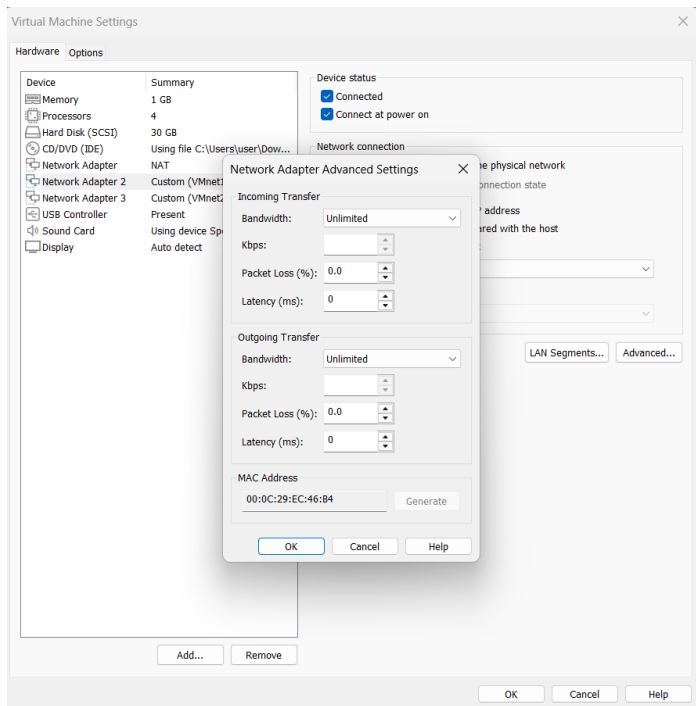
Interface	Network port	
WAN	em0 (00:50:56:2c:70:01)	
LAN	em1 (00:0c:29:ec:46:b4)	 Delete
OPT1	em2 (00:0c:29:ec:46:be)	 Delete

 Save

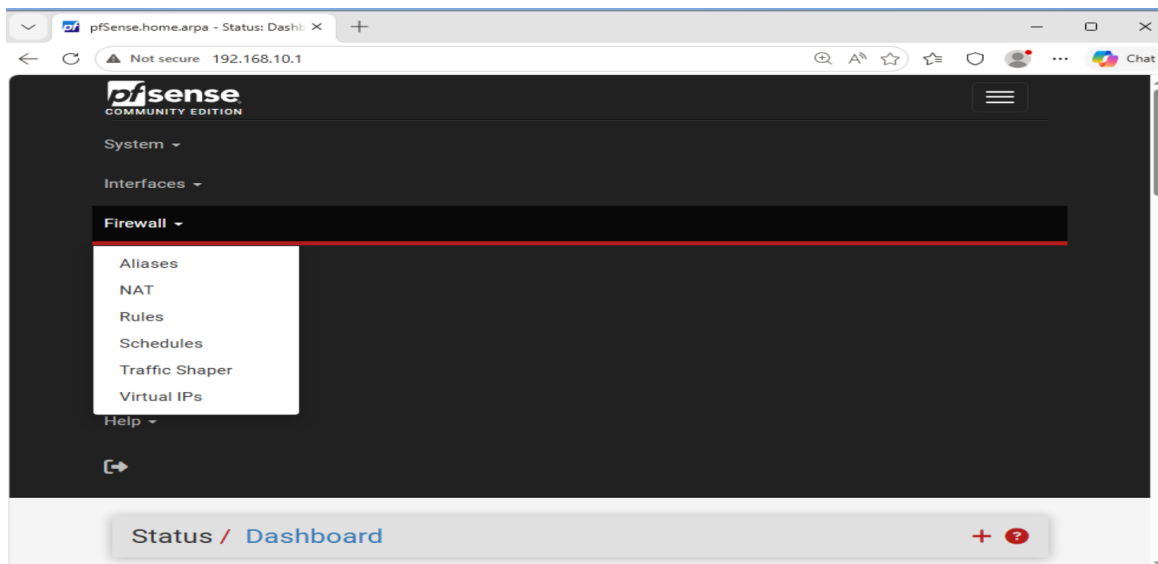
Firewall → Interfaces → Assignments



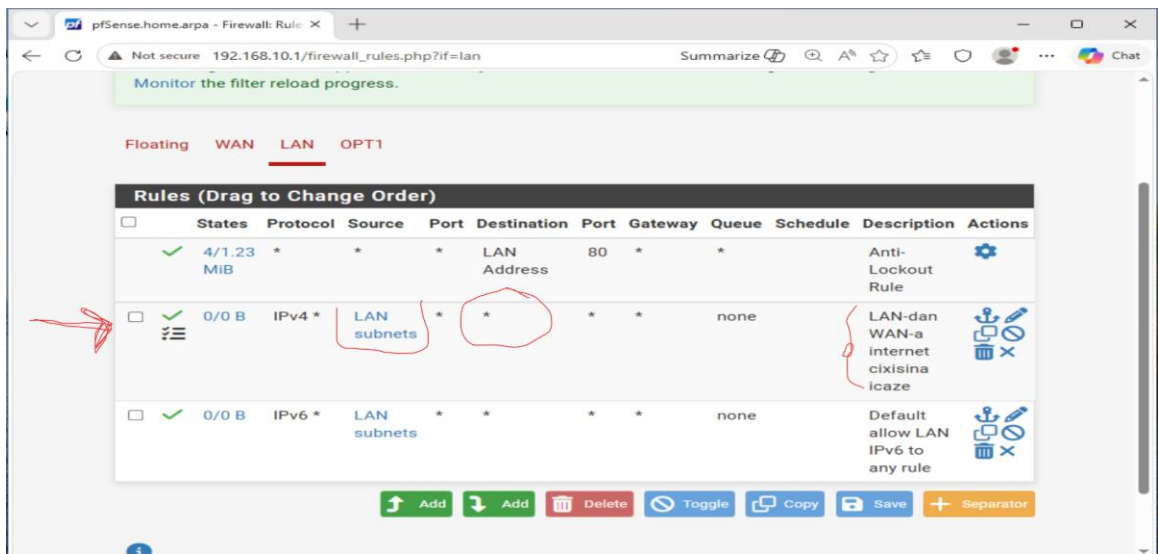
The MAC address for WAN is correct.



The MAC address for LAN is correct. The MAC address for OPT1 is correct.

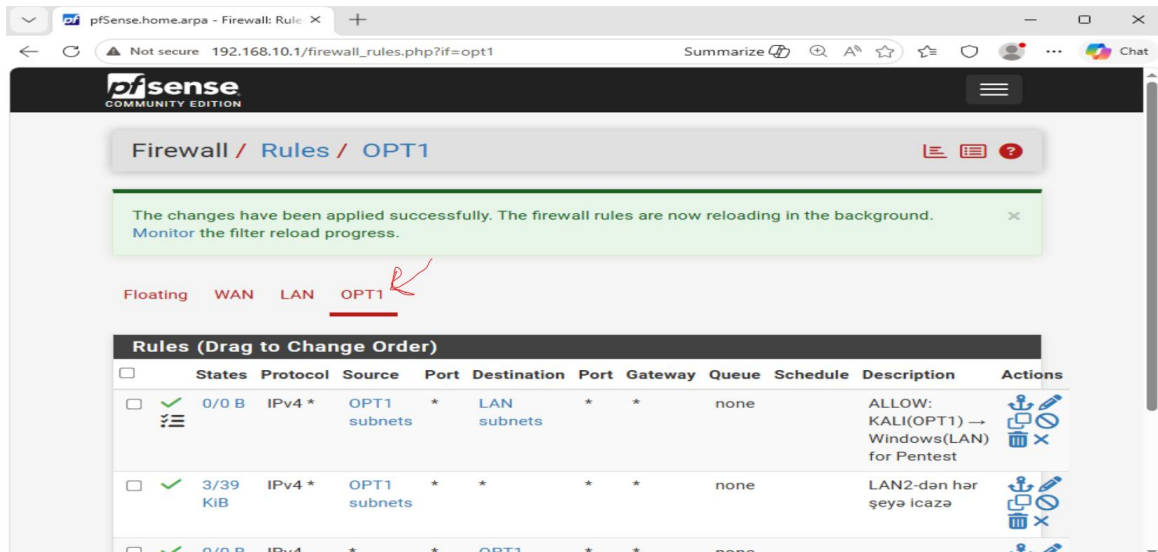


pfSense → Firewall → Rules



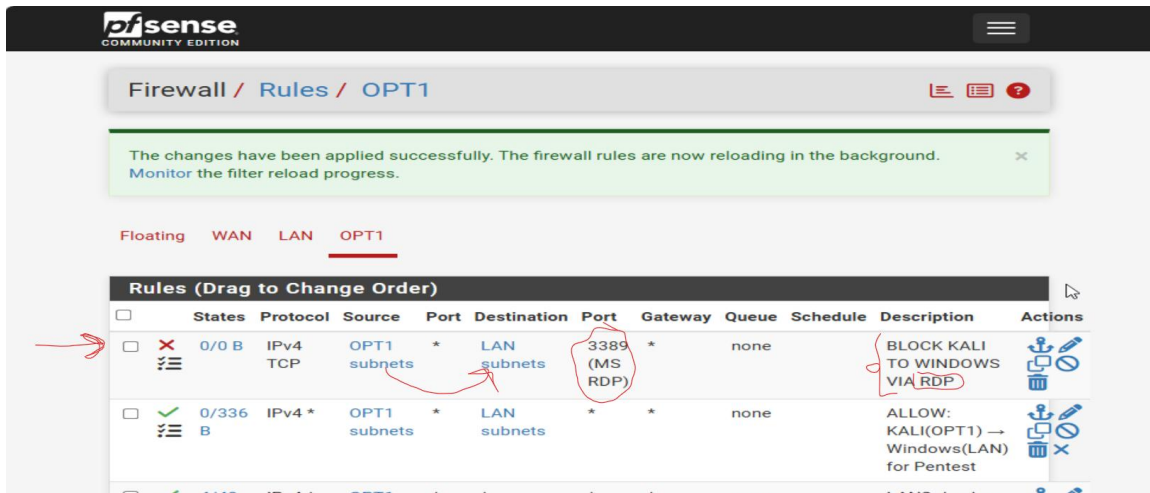
Rule added. Logging is also enabled for it.

7. Write a rule allowing traffic from OPT1 (Kali) to LAN (Windows) so the attack is possible (by default, OPT1 blocks everything).



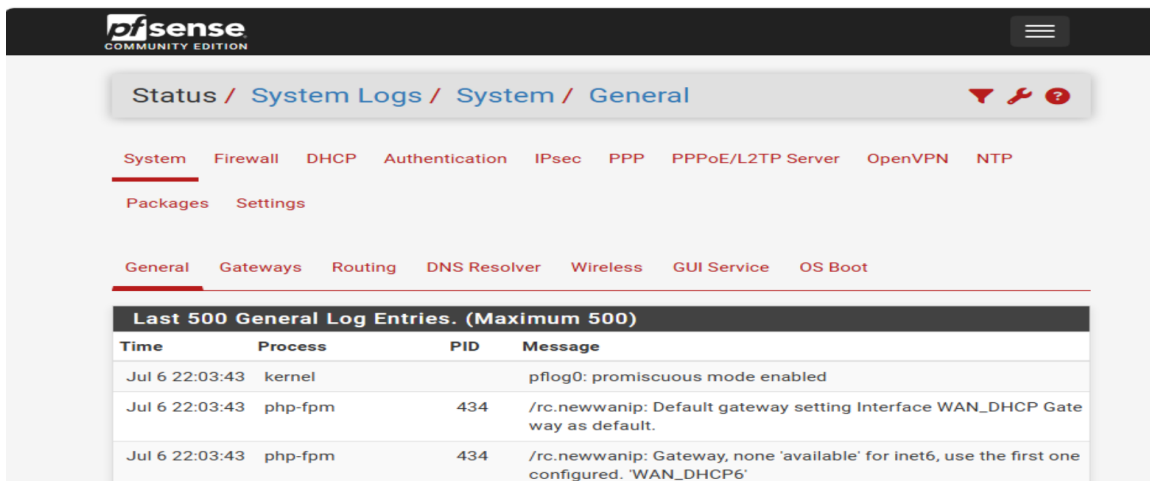
```
(kali@kali)-[~/Desktop]
$ ping 192.168.10.23
PING 192.168.10.23 (192.168.10.23) 56(84) bytes of data.
64 bytes from 192.168.10.23: icmp_seq=1 ttl=127 time=1.56 ms
64 bytes from 192.168.10.23: icmp_seq=2 ttl=127 time=1.51 ms
^C
— 192.168.10.23 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1113ms
rtt min/avg/max/mdev = 1.512/1.536/1.561/0.024 ms
(kali@kali)-[~/Desktop]
```

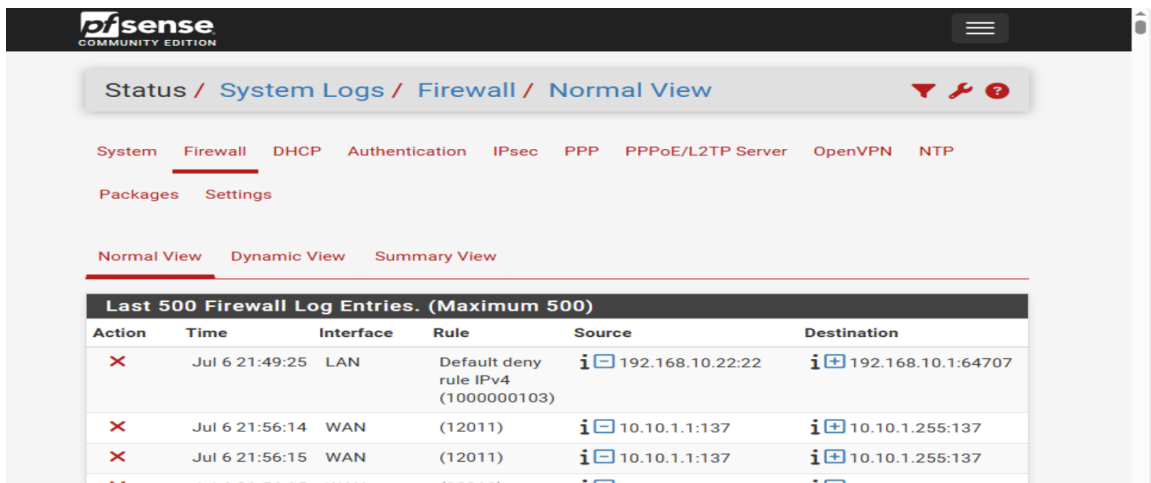
8. Create at least one blocking rule (e.g., block Kali's access to a specific port on Windows) and test it.



9. Make sure firewall logging is enabled (logs will later be sent to ELK).

pfSense → Status → System Logs



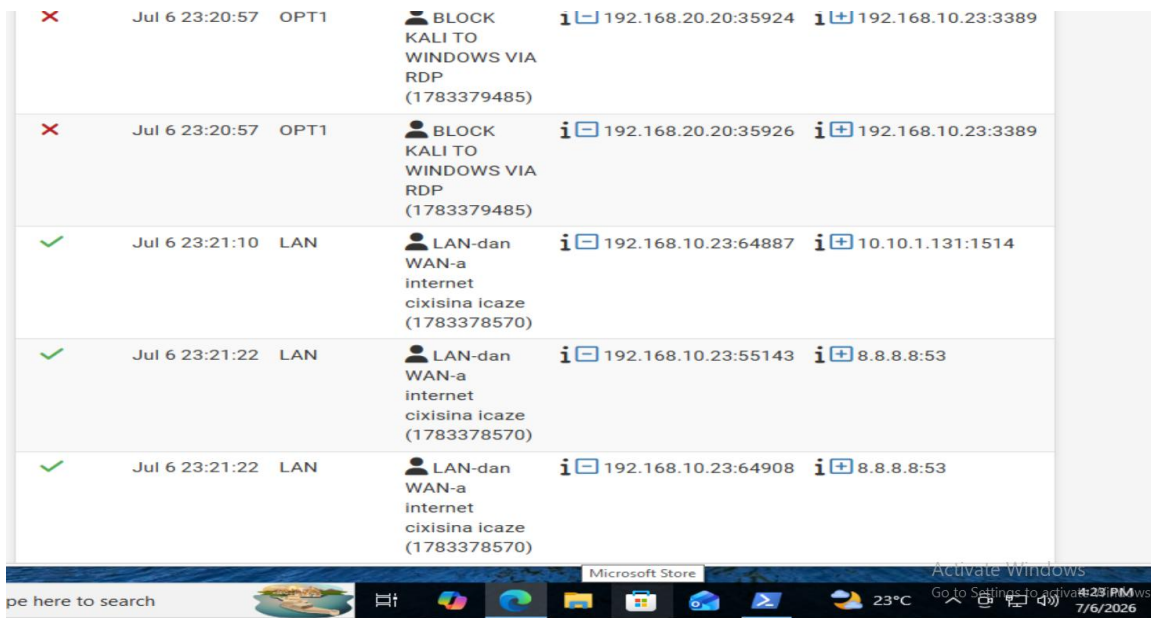


```
(kali@kali)-[~/Desktop]
$ nmap -p 3389 192.168.10.23
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-06 19:20 -0400
Nmap scan report for 192.168.10.23
Host is up (0.0022s latency).

PORT      STATE      SERVICE
3389/tcp  filtered  ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 0.90 seconds
```

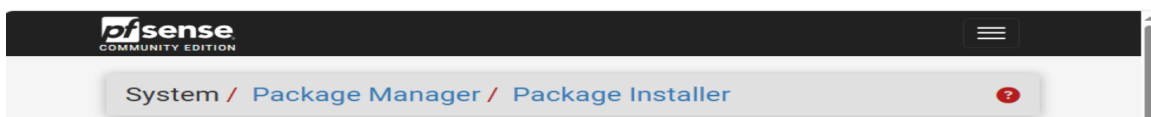
To test, I ran an NMAP scan against port 3389 — network mapping.



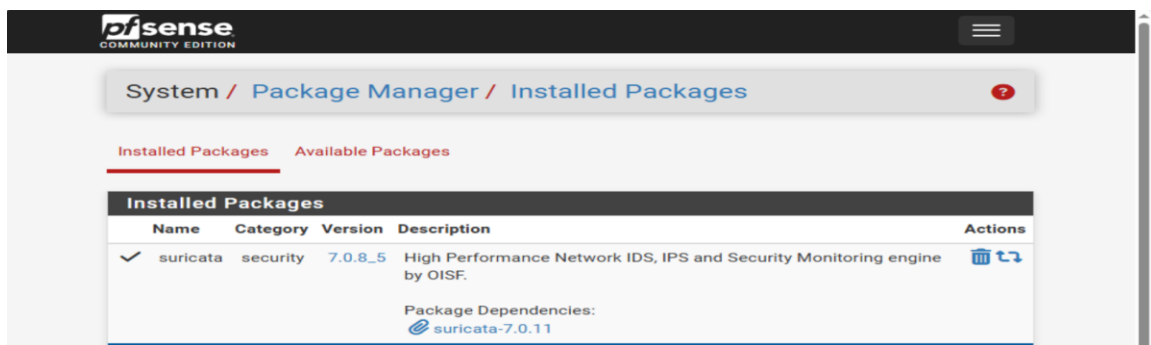
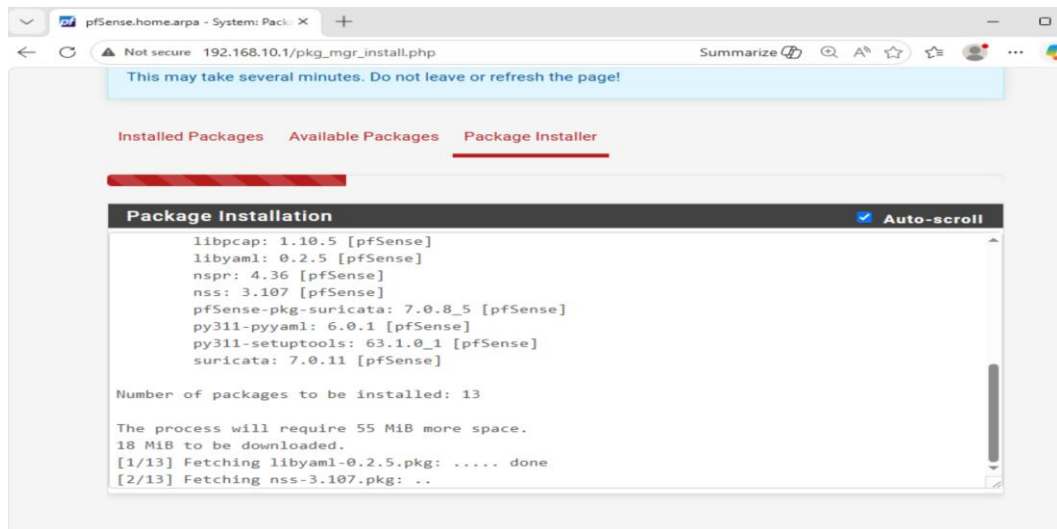
The event arrived as a log entry.

Phase 3 — IDS/IPS (Suricata)

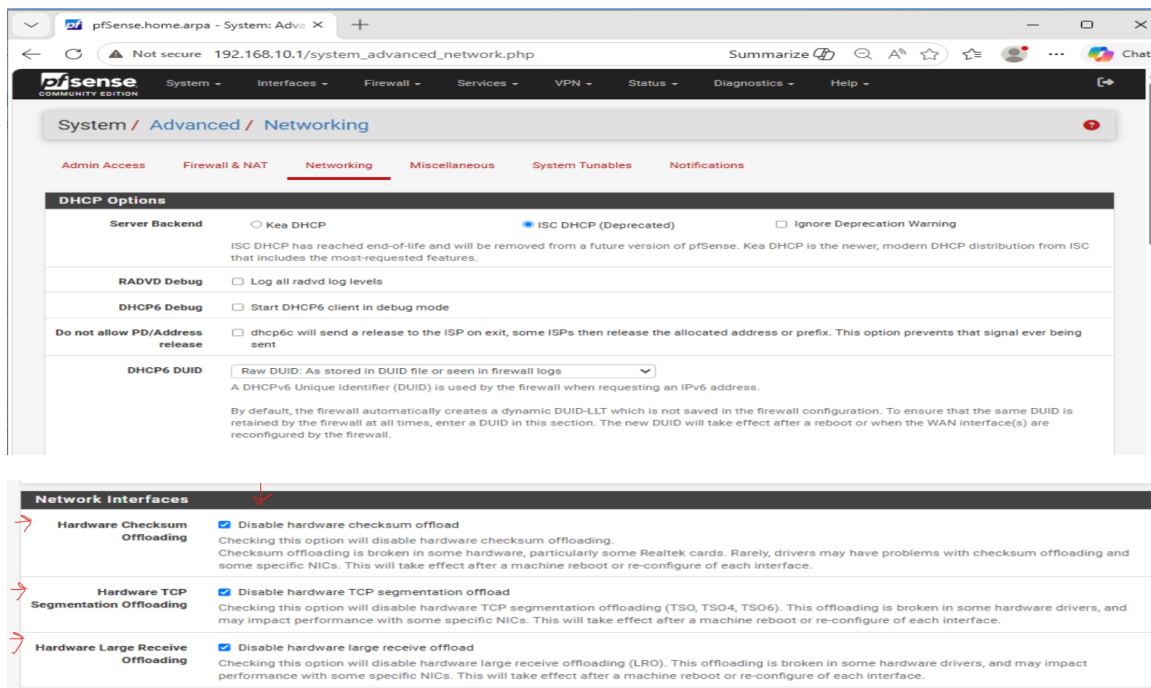
10. Install the Suricata package on pfSense and enable it on the LAN (or OPT1) interface so it can see the attack traffic.



pfSense → System → Package Manager: Installed Packages and Available Packages → Available Packages → Search term: suricata → +install → confirm

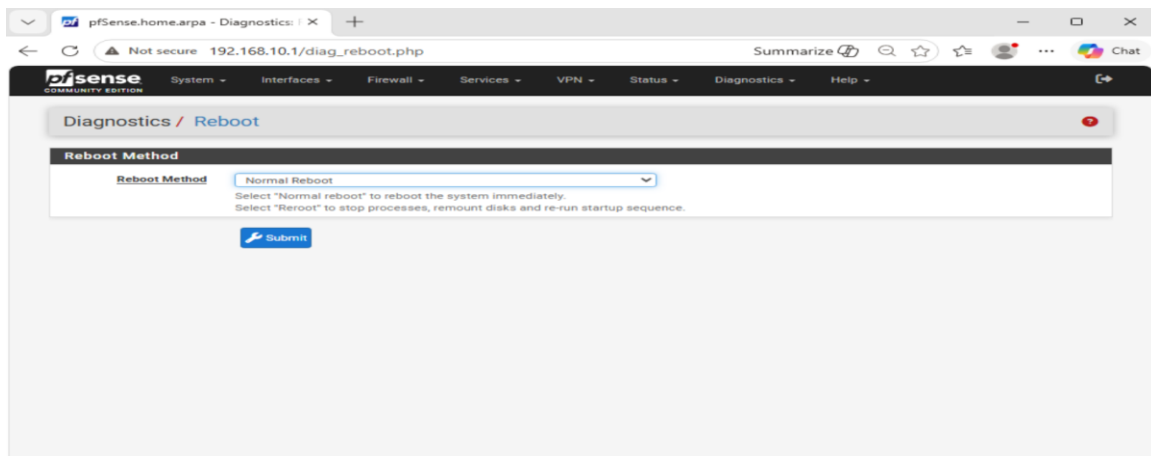


There was another issue:

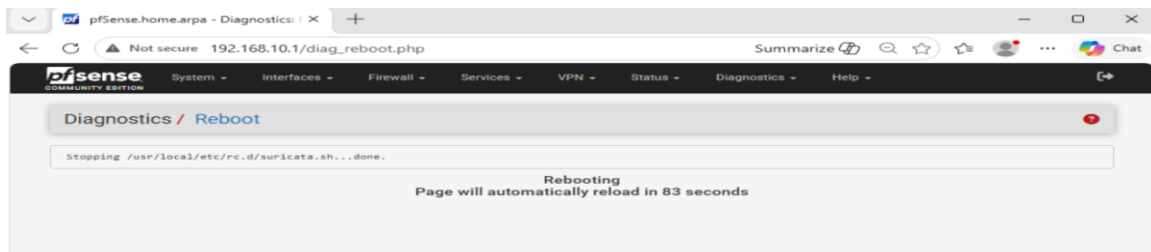


I disabled these settings so that Suricata itself — rather than the virtual adapter — could see and fully analyze the packets, allowing IDS/IPS to work correctly.

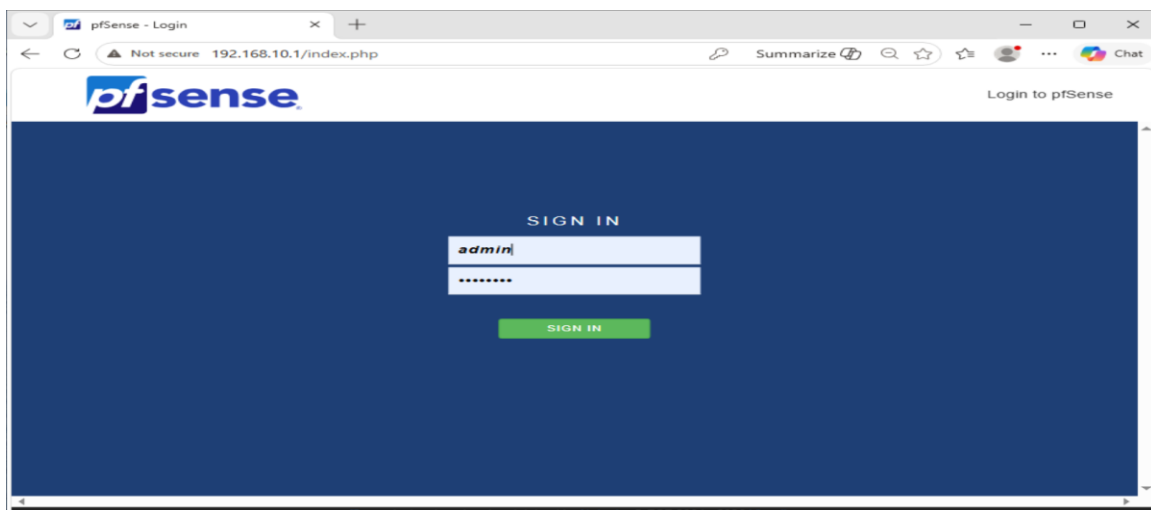
Saving triggers an automatic reboot prompt.



SUBMIT

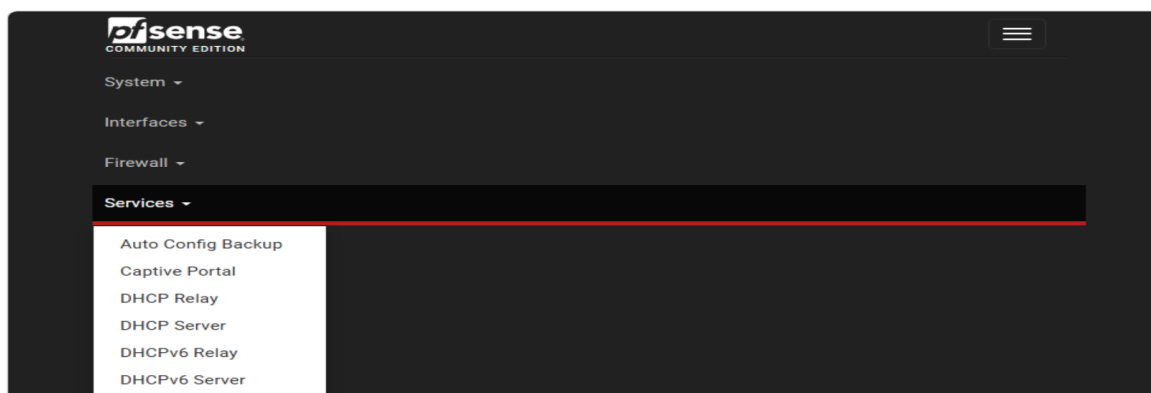


This screen confirms the process is running correctly.

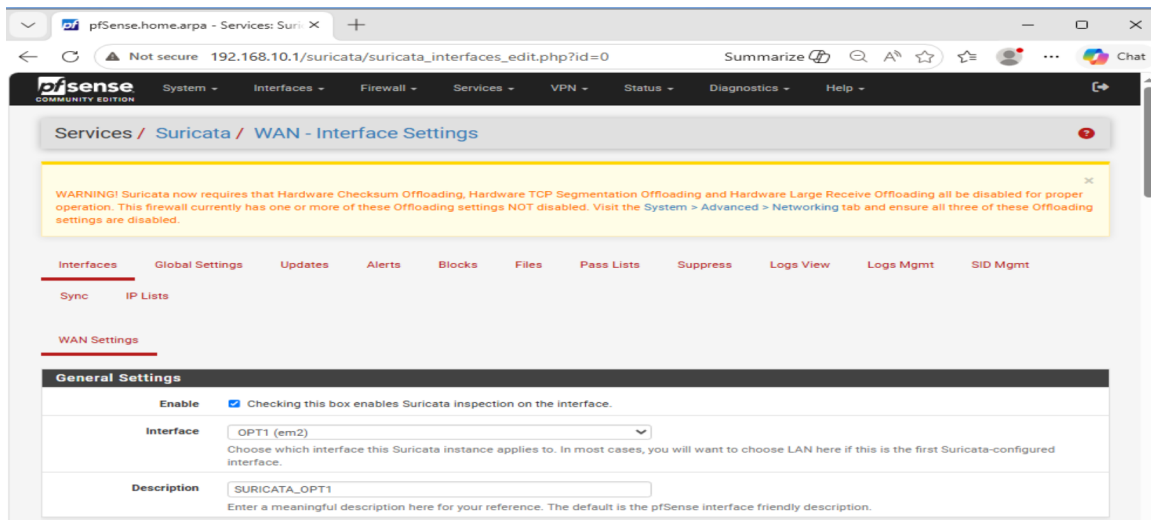
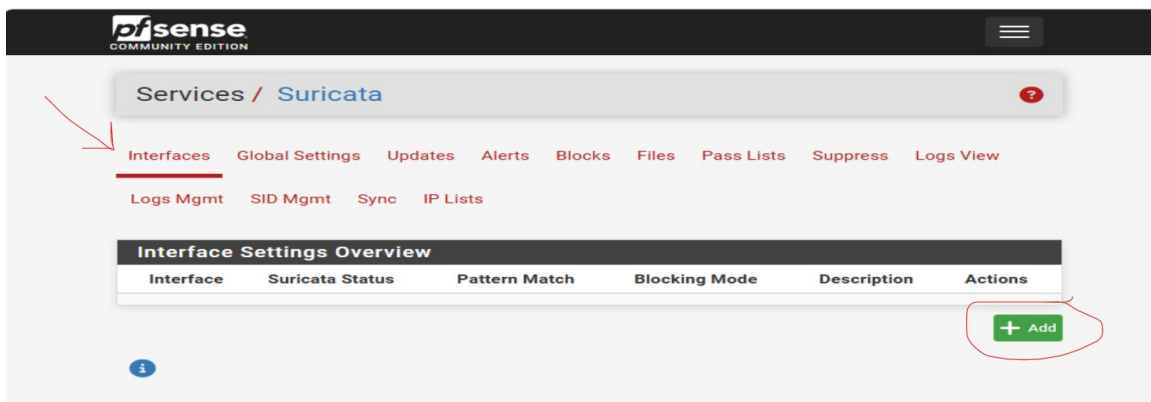


I re-enter the username and password and sign in.

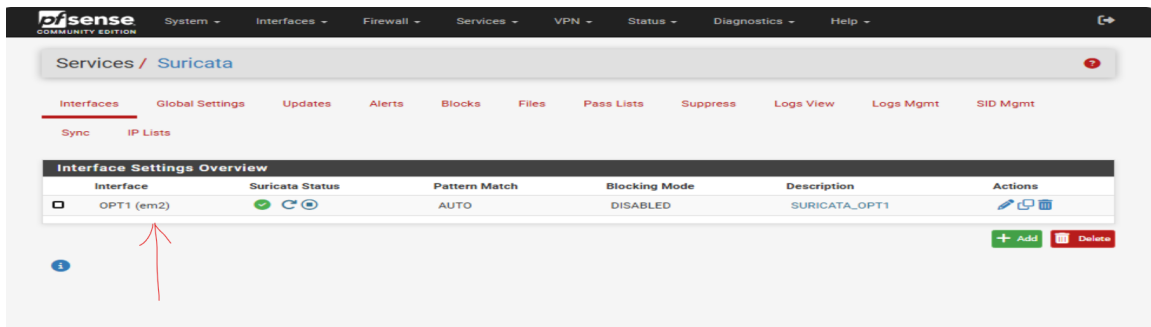
KALI (OPT1)



Services → suricata



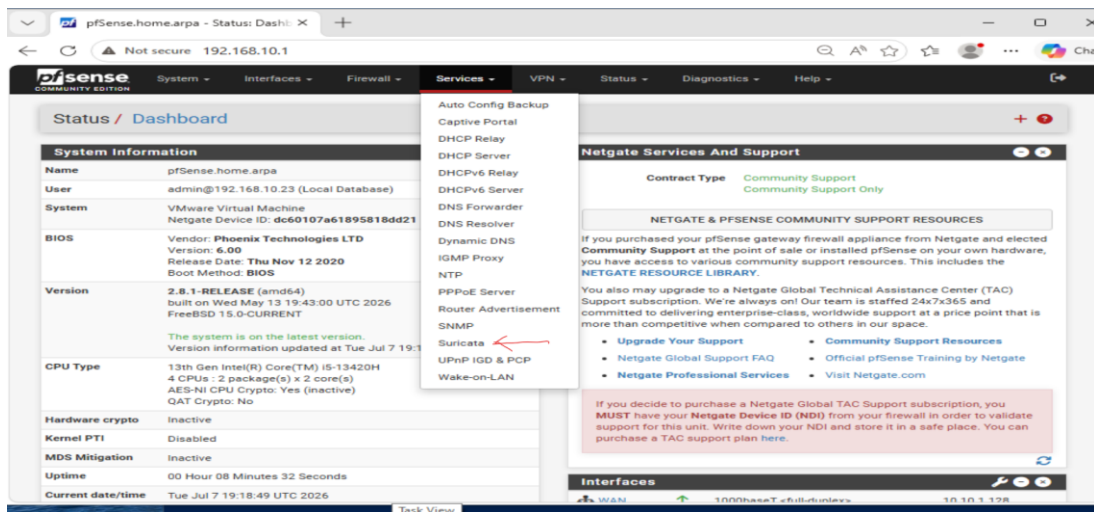
SAVE



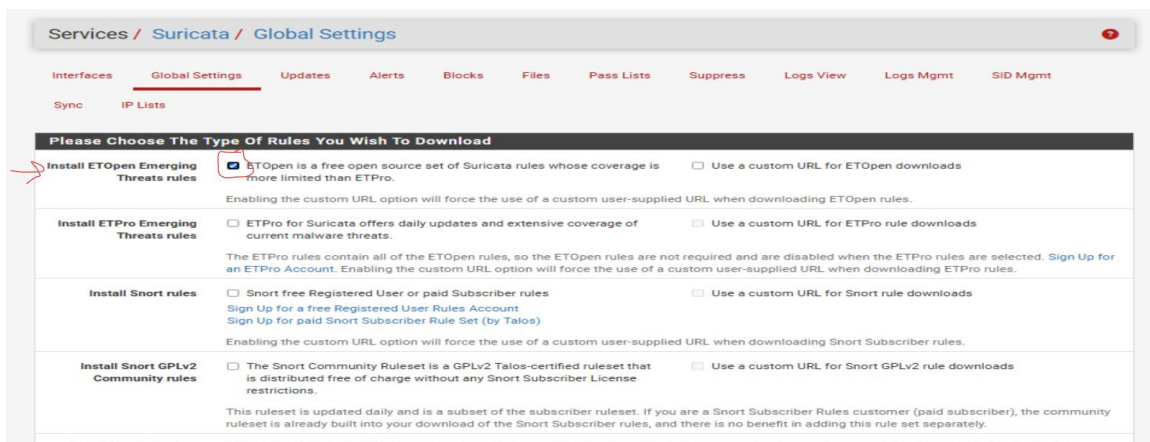
Suricata was enabled on OPT1 so that attack traffic coming from Kali, as it passes through pfSense, is monitored by Suricata. For now it is only logged as an alert and not blocked.

11. Download a ready-made ruleset (e.g., ET Open rules).

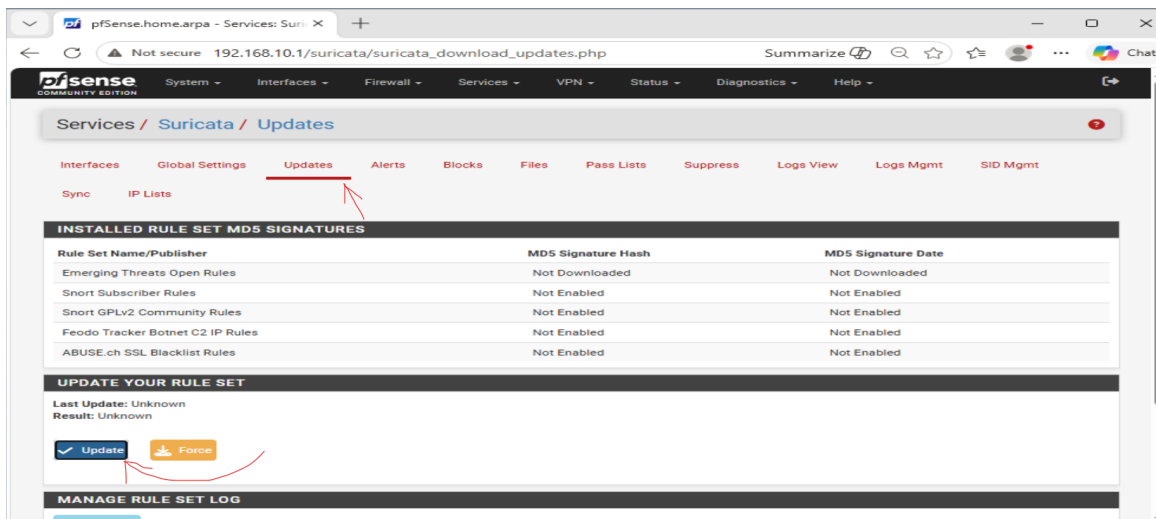
Services → Suricata



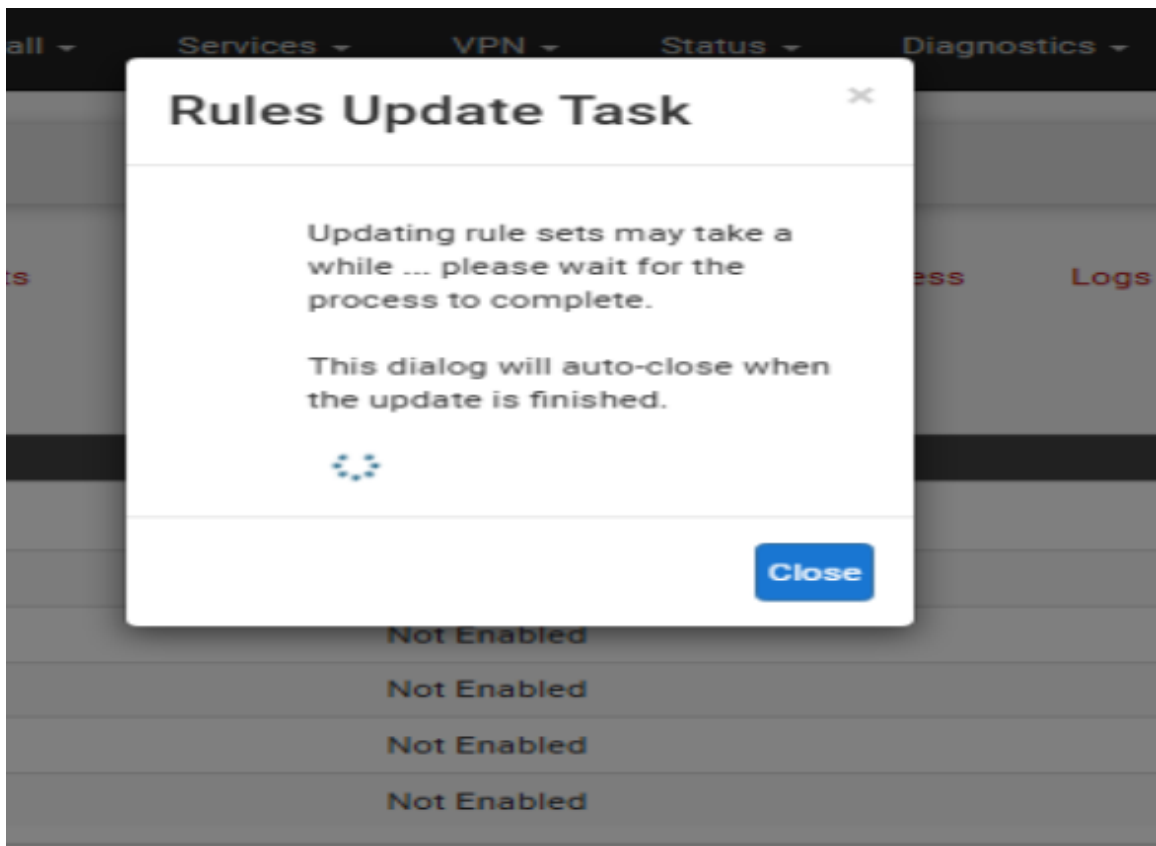
Suricata → Global Settings



SAVE



UPDATE

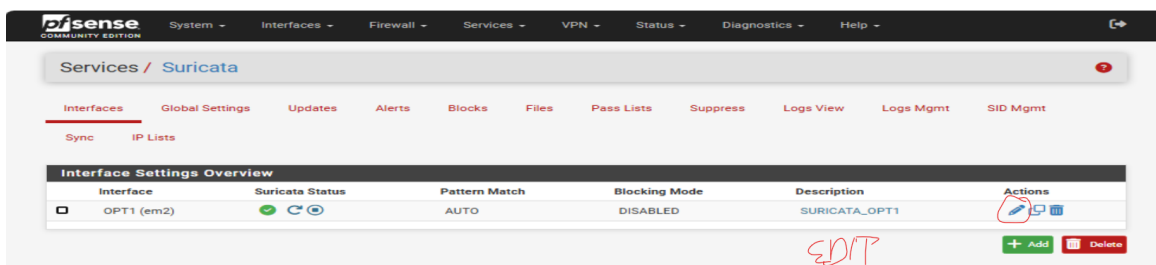


Waiting...

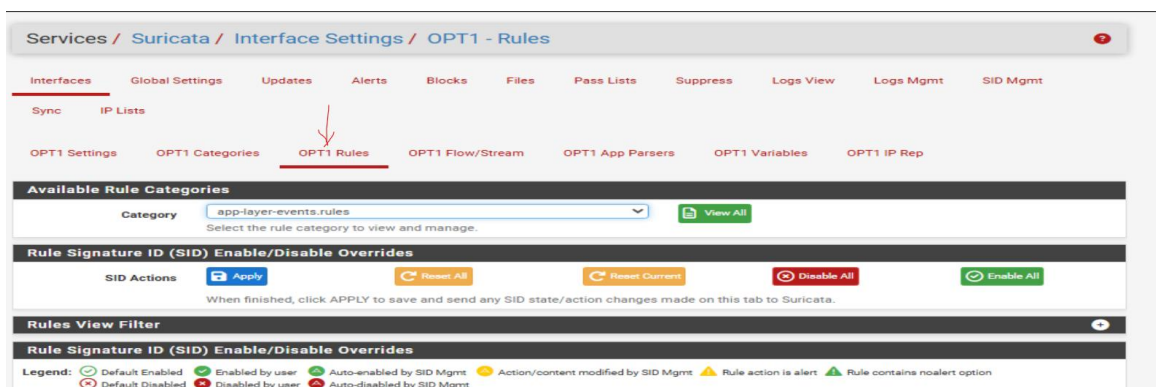
INSTALLED RULE SET MD5 SIGNATURES		
Rule Set Name/Publisher	MD5 Signature Hash	MD5 Signature Date
Emerging Threats Open Rules	8d4c421009e508beee86ac7ce093c488	Tuesday, 07-Jul-26 19:24:29 UTC
Snort Subscriber Rules	Not Enabled	Not Enabled
Snort GPLv2 Community Rules	Not Enabled	Not Enabled
Feodo Tracker Botnet C2 IP Rules	Not Enabled	Not Enabled
ABUSE.ch SSL Blacklist Rules	Not Enabled	Not Enabled

ET Open rules have now been downloaded.

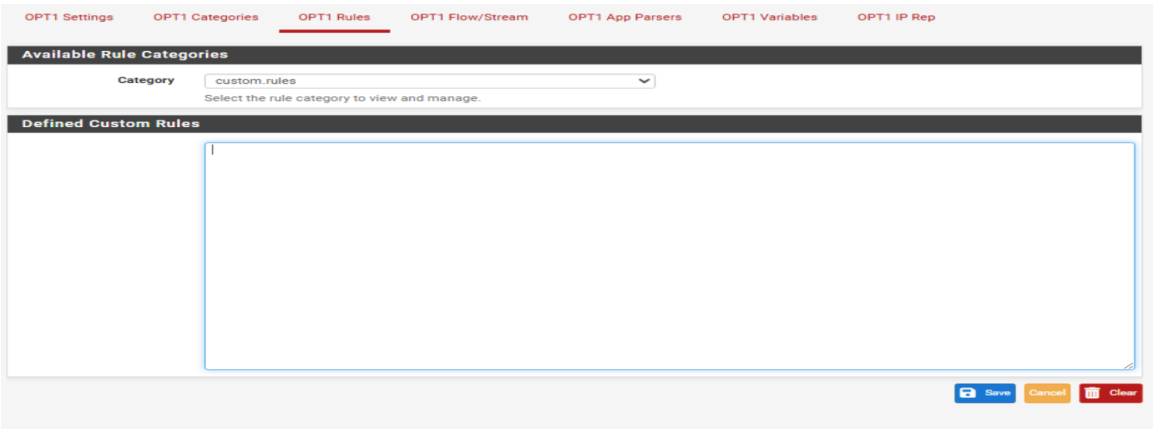
12. Write at least one custom rule — e.g., one that detects an nmap scan or a specific attack.



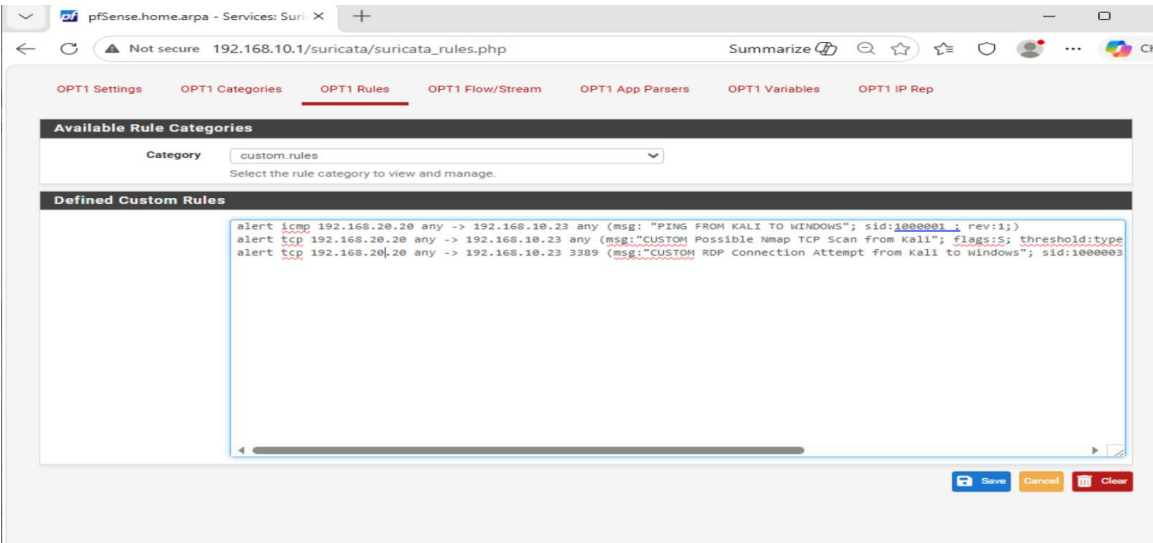
I start editing via the OPT1 interface.



From Interface Settings, I switch to OPT1 Rules.



I add the rule under Custom Logs.



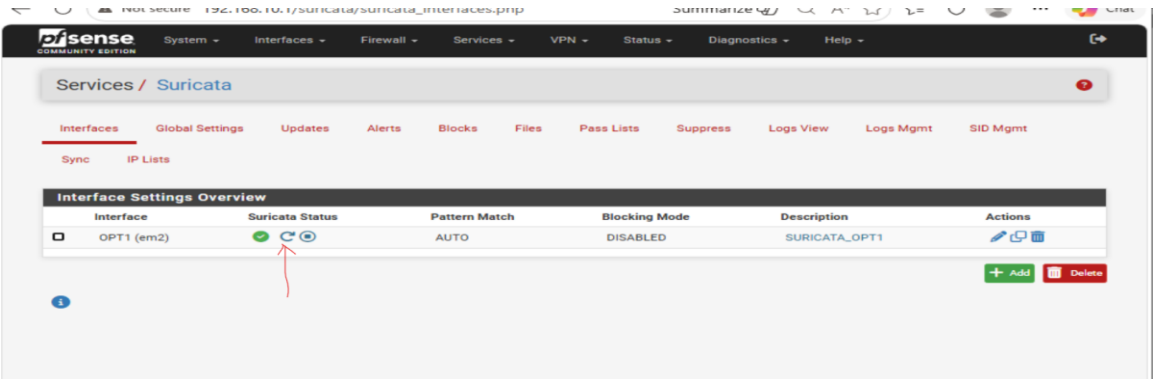
NOTE: the arrow here is not this symbol → — it is this one: ->

SAVE

```
alert icmp 192.168.20.20 any -> 192.168.10.23 any (msg:"Ping from KALI to WINDOWS";
sid:1000001; rev:1;)

alert tcp 192.168.20.20 any -> 192.168.10.23 any (msg:"CUSTOM Possible Nmap TCP Scan from
Kali"; flags:S; threshold:type both, track by_src, count 5, seconds 10; sid:1000002; rev:1;)

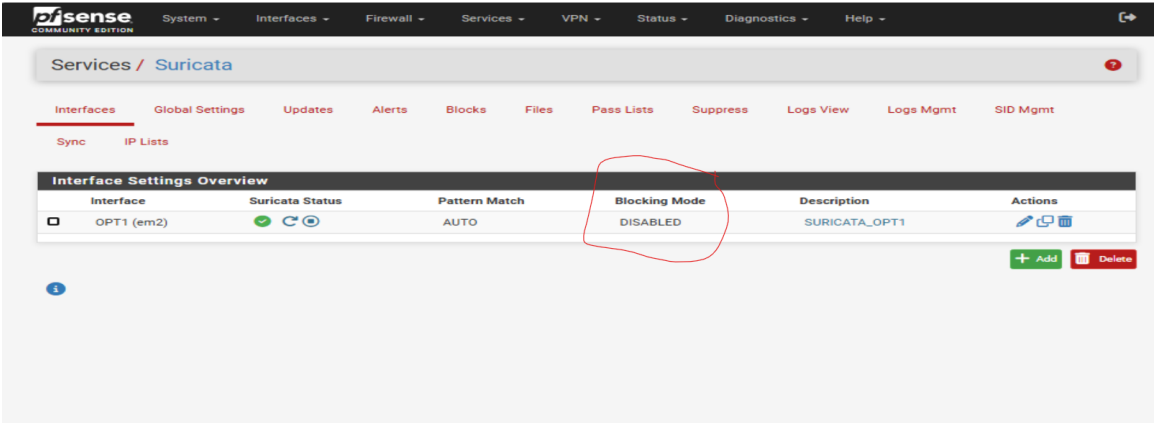
alert tcp 192.168.20.20 any -> 192.168.10.23 3389 (msg:"CUSTOM RDP Connection Attempt from
Kali to Windows"; sid:1000003; rev:1;)
```



RESTART

After writing the custom rule, Suricata on OPT1 is restarted so the new rule takes effect. (Green status light)

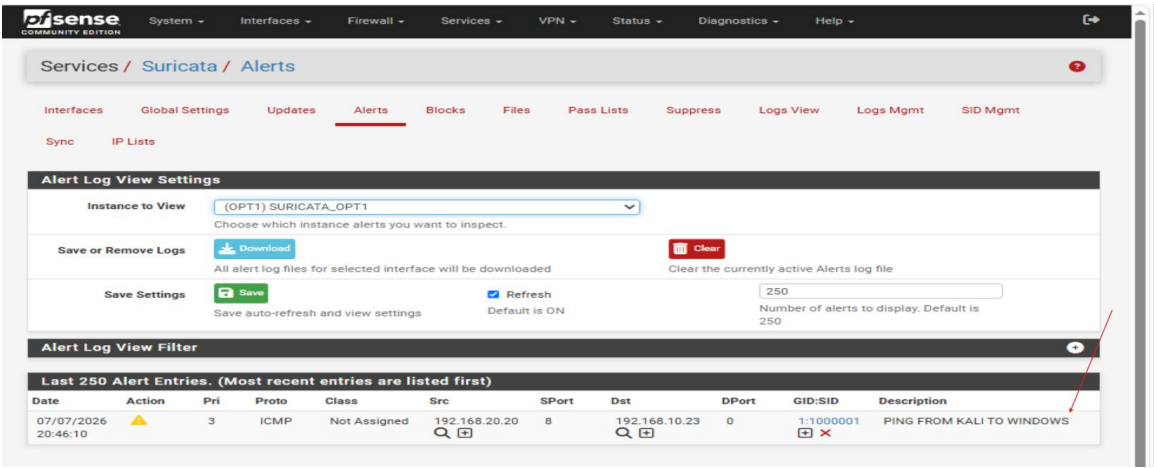
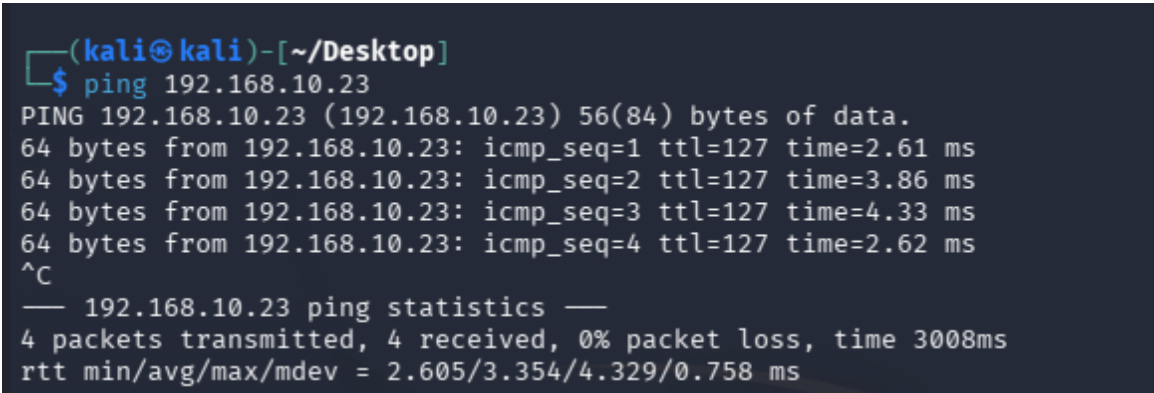
13. First test in IDS (detection-only) mode, then switch to inline IPS (blocking) mode and show the difference.



This confirms IDS mode is active. In short, Suricata generates an alert but does not block — the traffic still passes through.

Let's check (IDS):

[1]



An alert was generated by pinging from KALI to Windows.

[2]

```
(kali@kali)-[~/Desktop]
$ nmap 192.168.10.23
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-07 16:48 -0400
Nmap scan report for 192.168.10.23
Host is up (0.0037s latency).
All 1000 scanned ports on 192.168.10.23 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 21.93 seconds
```

pfSense.home.arpa - Services: Suricata

192.168.10.1/suricata/suricata_alerts.php

Alert Log View Settings

Instance to View: (OPT1) SURICATA_OPT1

Save or Remove Logs: Download Clear

Save Settings: Save Refresh Default is ON

Number of alerts to display: 250

Alert Log View Filter

Last 250 Alert Entries. (Most recent entries are listed first)

Date	Action	Pri	Proto	Class	Src	SPort	Dst	DPort	GID/SID	Description
07/07/2026 20:48:15	Warning	3	TCP	Not Assigned	192.168.20.20	62656	192.168.10.23	3389	1:1000003	CUSTOM RDP Connection Attempt from Kali to Windows
07/07/2026 20:48:15	Warning	3	TCP	Not Assigned	192.168.20.20	62654	192.168.10.23	3389	1:1000003	CUSTOM RDP Connection Attempt from Kali to Windows
07/07/2026 20:48:13	Warning	3	TCP	Not Assigned	192.168.20.20	62654	192.168.10.23	23	1:1000002	CUSTOM Possible Nmap TCP Scan from Kali
07/07/2026 20:48:13	Warning	3	ICMP	Not Assigned	192.168.20.20	13	192.168.10.23	0	1:1000001	PING FROM KALI TO WINDOWS
07/07/2026 20:46:10	Warning	3	ICMP	Not Assigned	192.168.20.20	8	192.168.10.23	0	1:1000001	PING FROM KALI TO WINDOWS

Nmap reported the ports as "filtered" because it received no response — but Suricata saw the scan traffic and generated an alert based on the custom rule.

In IPS mode, it both detects and blocks.

pfSense COMMUNITY EDITION

System - Interfaces - Firewall - Services - VPN - Status - Diagnostics - Help

Services / Suricata

Interfaces Global Settings Updates Alerts Blocks Files Pass Lists Suppress Logs View Logs Mgmt SID Mgmt

Sync IP Lists

Interface Settings Overview

Interface	Suricata Status	Pattern Match	Blocking Mode	Description	Actions
OPT1 (em2)	Enabled	AUTO	DISABLED	SURICATA_OPT1	Edit Add Delete

EDIT

I edit it.

Alert and Block Settings

Block Offenders: ☒ Checking this option will automatically block hosts that generate a Suricata alert.

IPS Mode:

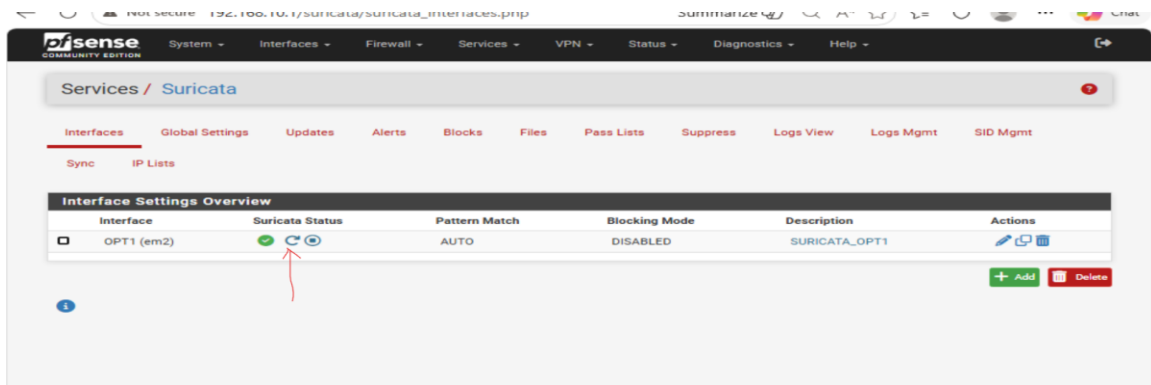
Select blocking mode operation. Legacy Mode inspects copies of packets while Inline Mode inserts the Suricata inspection engine into the network stack between the NIC and the OS. Default is Legacy Mode.

Legacy Mode uses the PCAP engine to generate copies of packets for inspection as they traverse the interface. Some "leakage" of packets will occur before Suricata can determine if the traffic matches a rule and should be blocked. Inline mode instead intercepts and inspects packets before they are handed off to the host network stack for further processing. Packets matching DROP rules are simply discarded (dropped) and not passed to the host network stack. No leakage of packets occurs with Inline Mode. WARNING: Inline Mode only works with NIC drivers which properly support Netmap! Supported drivers include: bnxt, cc, cxgbe, cxl, em, ens, ice, igb, igc, ix, ixgbe, ixl, lem, re, vmx, vtnet. If problems are experienced with Inline Mode, switch to Legacy Mode instead.

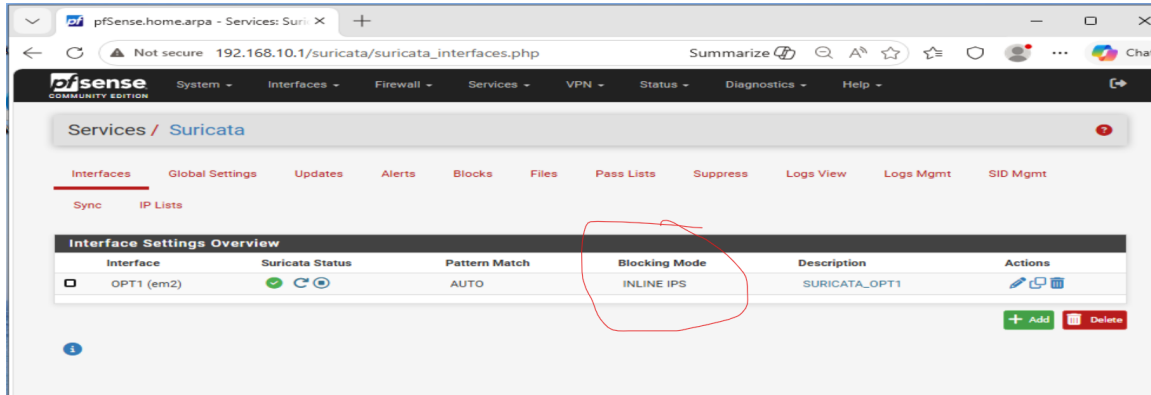
Netmap Threads:

Enter the number of netmap threads to use. Default is "auto" and is recommended. When set to "auto", Suricata will query the system for the number of supported netmap queues, and it will use a matching number of netmap threads. The NIC hosting this interface registered 1 queue(s) with the kernel.

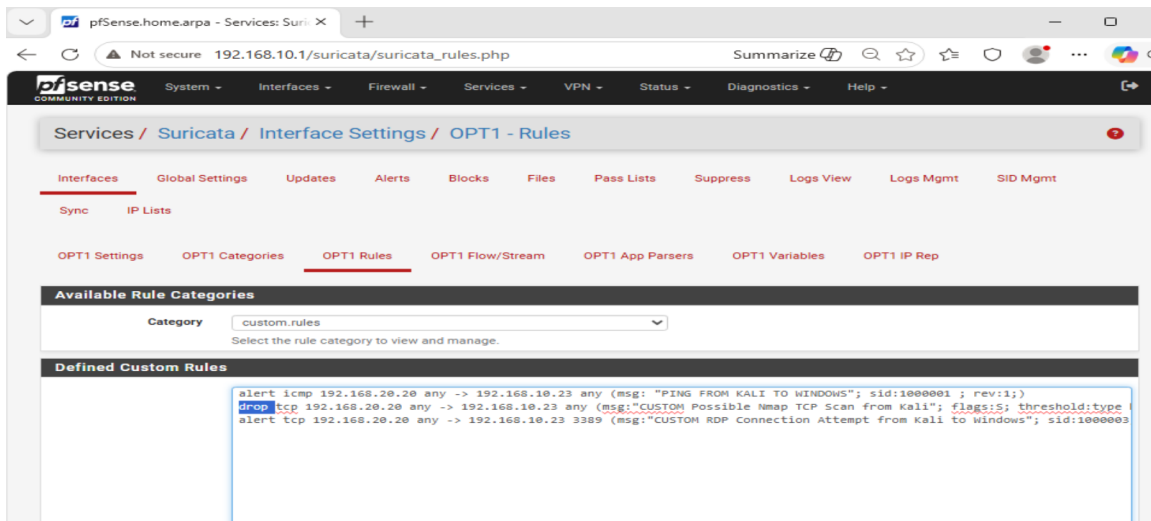
IPS Mode: Inline Mode → enabled. SAVE



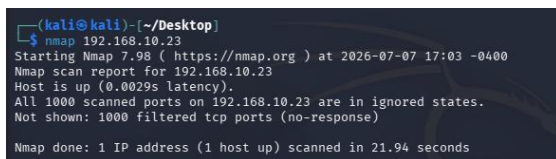
RESTART



I make a small change to the custom rules so they don't just alert, but also drop.



Let's check:

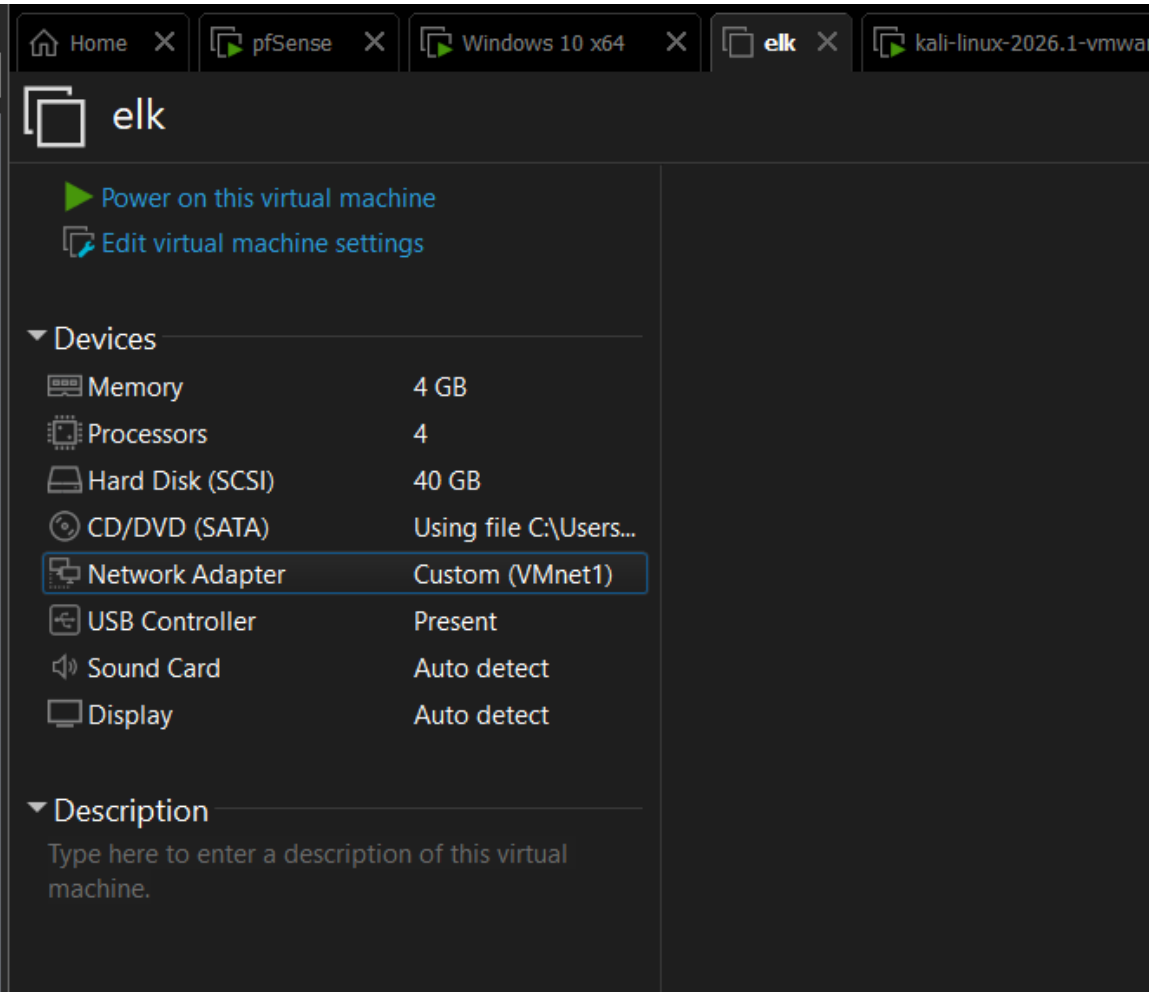


Yellow triangle = acting as IDS — it only generated an alert; traffic still passed through.
Red icon / red text = acting as IPS — it generated an alert AND dropped/blocked the packet.

This means the packets never reach the target and no response comes back.

Phase 4 — SIEM (ELK Stack)

14. Set up the ELK server (Elasticsearch + Kibana).




```

Ubuntu 26.04 LTS elk tty1

Try contacting this VM's SSH server via 'ssh vsock%3606578072' from host.

elk login: elk
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-22-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jul  7 09:51:56 PM UTC 2026

System load:  0.54           Processes:    272
Usage of /:   58.9% of 18.53GB Users logged in:  0
Memory usage: 65%          IPv4 address for ens33: 192.168.10.22
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

35 updates can be applied immediately.
26 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

elk@elk:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:f8:03:98 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29f80398
    inet 192.168.10.22/24 metric 100 brd 192.168.10.255 scope global dynamic ens33
        valid_lft 7186sec preferred_lft 7186sec
    inet6 fe80::20c:29ff:fe8:398/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
elk@elk:~$ _

```

```

elk@elk:~$ sudo systemctl status kibana
* kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; disabled; preset: enabled)
   Active: inactive (dead)
   Docs: https://www.elastic.co
elk@elk:~$ _

```

```

elk@elk:~$ sudo systemctl start kibana
elk@elk:~$ sudo systemctl status kibana
* kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; disabled; preset: enabled)
   Active: active (running) since Tue 2026-07-07 22:02:17 UTC; 2s ago
   Invocation: 540878c737de4c12ab31daa6dbab00dd
   Docs: https://www.elastic.co
   Main PID: 2617 (MainThread)
   Tasks: 7 (limit: 3937)
   Memory: 177.1M (peak: 177.1M)
   CPU: 2.448s
   CGroup: /system.slice/kibana.service
           └─2617 /usr/share/kibana/bin/./node/default/bin/node /usr/share/kibana/bin/./src/cli/kibana/dist

Jul 07 22:02:17 elk systemd[1]: Started kibana.service - Kibana.
elk@elk:~$ _

```

```

elk@elk:~$ sudo systemctl status elasticsearch
* elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-07-07 21:52:25 UTC; 11min ago
   Invocation: e321f5c67a4c29e999dbf5909109
   Docs: https://www.elastic.co
   Main PID: 1997 (server-launcher)
   Tasks: 97 (limit: 3937)
   Memory: 1.2G (peak: 2.4G, swap: 1G, swap peak: 1G)
   CPU: 1min 59.94s
   CGroup: /system.slice/elasticsearch.service
           └─1997 /usr/share/elasticsearch/lib/tools/server-launcher/server-launcher -p /var/run/elasticsearch/elasticsearch.pid --quiet
              2072 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.negative.ttl=10 -XX:+AlwaysPreTouch -Xss10
              2131 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bin/controller

Jul 07 21:51:47 elk systemd[1]: Starting elasticsearch.service - Elasticsearch...
Jul 07 21:52:16 elk systemd-entrypoint[1997]: WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
Jul 07 21:52:16 elk systemd-entrypoint[1997]: WARNING: sun.misc.Unsafe:arrayBaseOffset has been called by com.google.protobuf.UnsafeUtil$MemoryAccessor (file:
Jul 07 21:52:16 elk systemd-entrypoint[1997]: WARNING: Please consider reporting this to the maintainers of class com.google.protobuf.UnsafeUtil$MemoryAccessor
Jul 07 21:52:16 elk systemd-entrypoint[1997]: WARNING: sun.misc.Unsafe:arrayBaseOffset will be removed in a future release
Jul 07 21:52:25 elk systemd[1]: Started elasticsearch.service - Elasticsearch.

```

Activate Windows

Elasticsearch and Kibana are active, but browsing to <http://192.168.10.22:5601> doesn't open it. Running "sudo ss -tupln | grep 5601" should show 0.0.0.0:5601.

```
elk@elk:~$ sudo nano /etc/elasticsearch/elasticsearch.yml
```

```
# Enable security features
xpack.security.enabled: true

xpack.security.enrollment.enabled: true

# Enable encryption for HTTP API client connections, such as Kibana, Logstash, and Agents
xpack.security.http.ssl:
  enabled: true
  keystore.path: certs/http.p12

# Enable encryption and mutual authentication between cluster nodes
xpack.security.transport.ssl:
  enabled: true
  verification_mode: certificate
  keystore.path: certs/transport.p12
  truststore.path: certs/transport.p12
# Create a new cluster with the current node only
# Additional nodes can still join the cluster later
cluster.initial_master_nodes: ['elk']

# Allow HTTP API connections from anywhere
# Connections are encrypted and require user authentication
http.host: 0.0.0.0

# Allow other nodes to join the cluster from anywhere
# Connections are encrypted and mutually authenticated
#transport.host: 0.0.0.0
```

I replaced the "true" values here with "false".

```
# Enable security features
xpack.security.enabled: false

xpack.security.enrollment.enabled: false

# Enable encryption for HTTP API client connections, such as Kibana, Logstash, and Agents
xpack.security.http.ssl:
  enabled: false
  keystore.path: certs/http.p12

# Enable encryption and mutual authentication between cluster nodes
xpack.security.transport.ssl:
  enabled: false
  verification_mode: certificate
  keystore.path: certs/transport.p12
  truststore.path: certs/transport.p12
# Create a new cluster with the current node only
# Additional nodes can still join the cluster later
cluster.initial_master_nodes: ['elk']

# Allow HTTP API connections from anywhere
# Connections are encrypted and require user authentication
http.host: 0.0.0.0

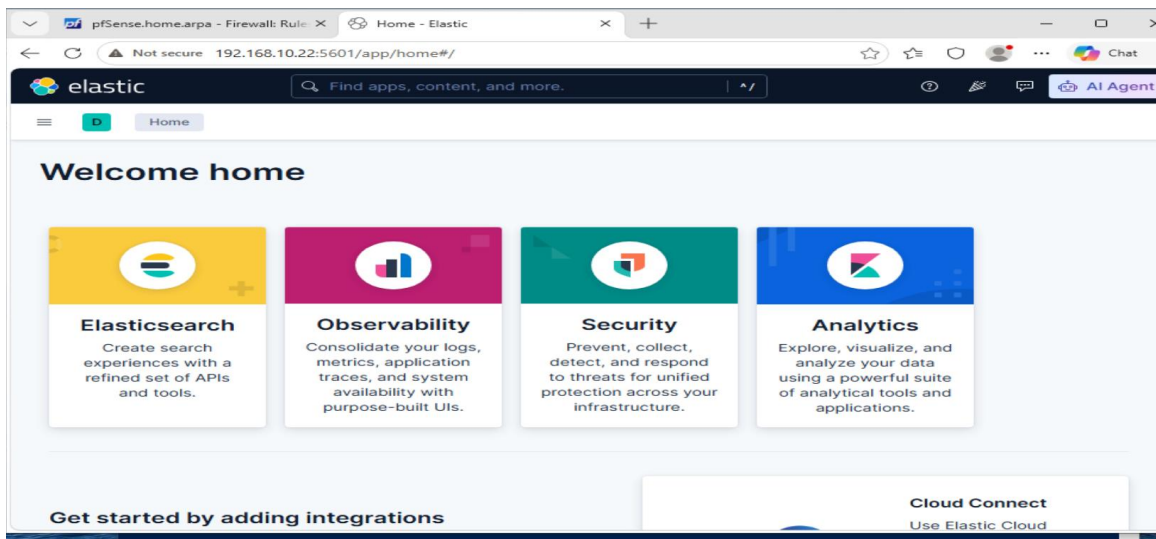
# Allow other nodes to join the cluster from anywhere
# Connections are encrypted and mutually authenticated
#transport.host: 0.0.0.0

#----- END SECURITY AUTO CONFIGURATION -----
```

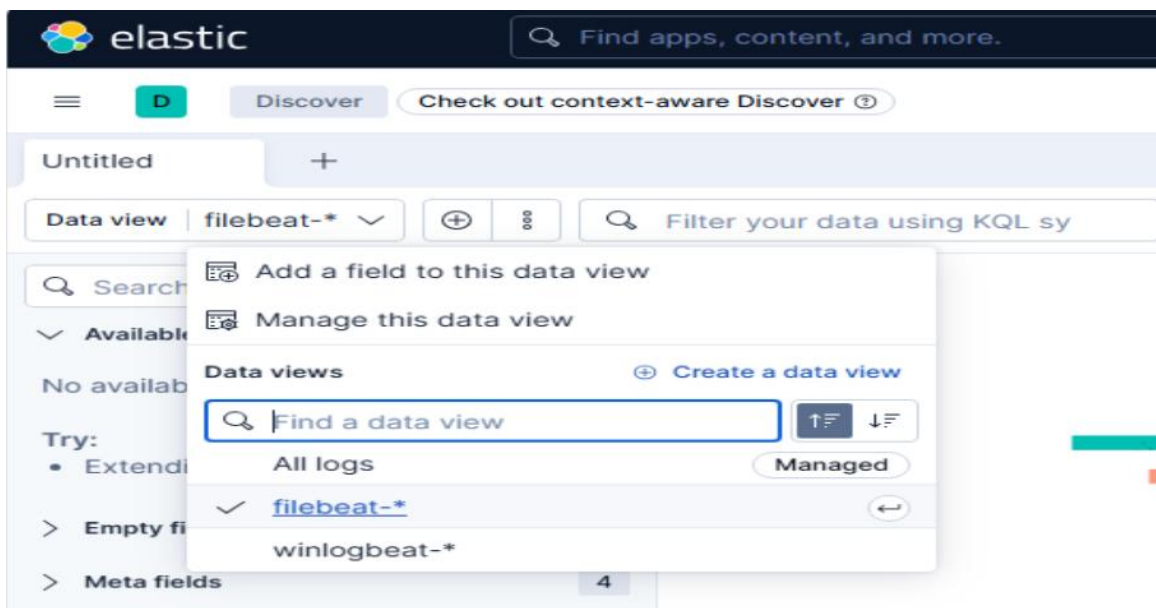
I set security to false so that, in this lab environment, Kibana can connect to Elasticsearch over plain <http://localhost:9200> without requiring authentication or an HTTPS certificate.

CTRL+O → ENTER → CTRL+X

```
^C
elk@elk:~$ sudo systemctl restart elasticsearch
```

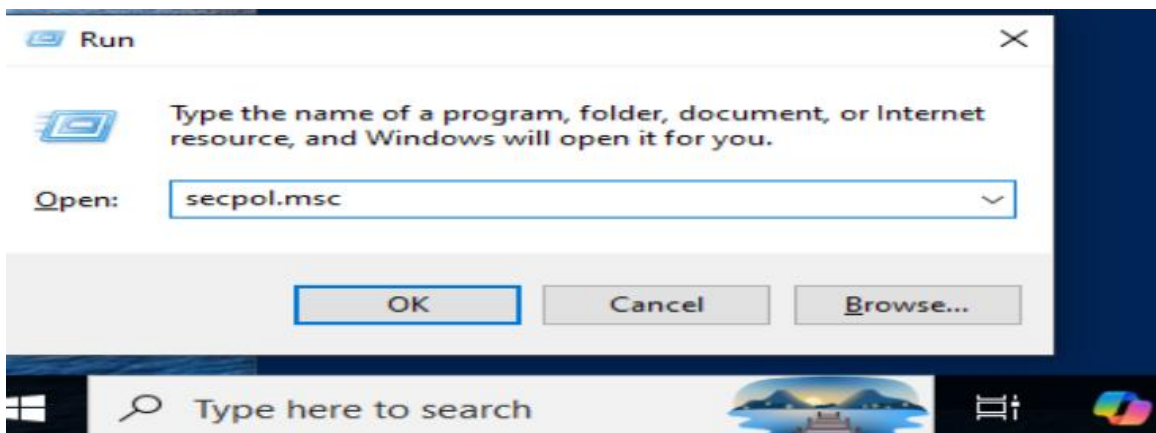


Let's check the agents in Discover.

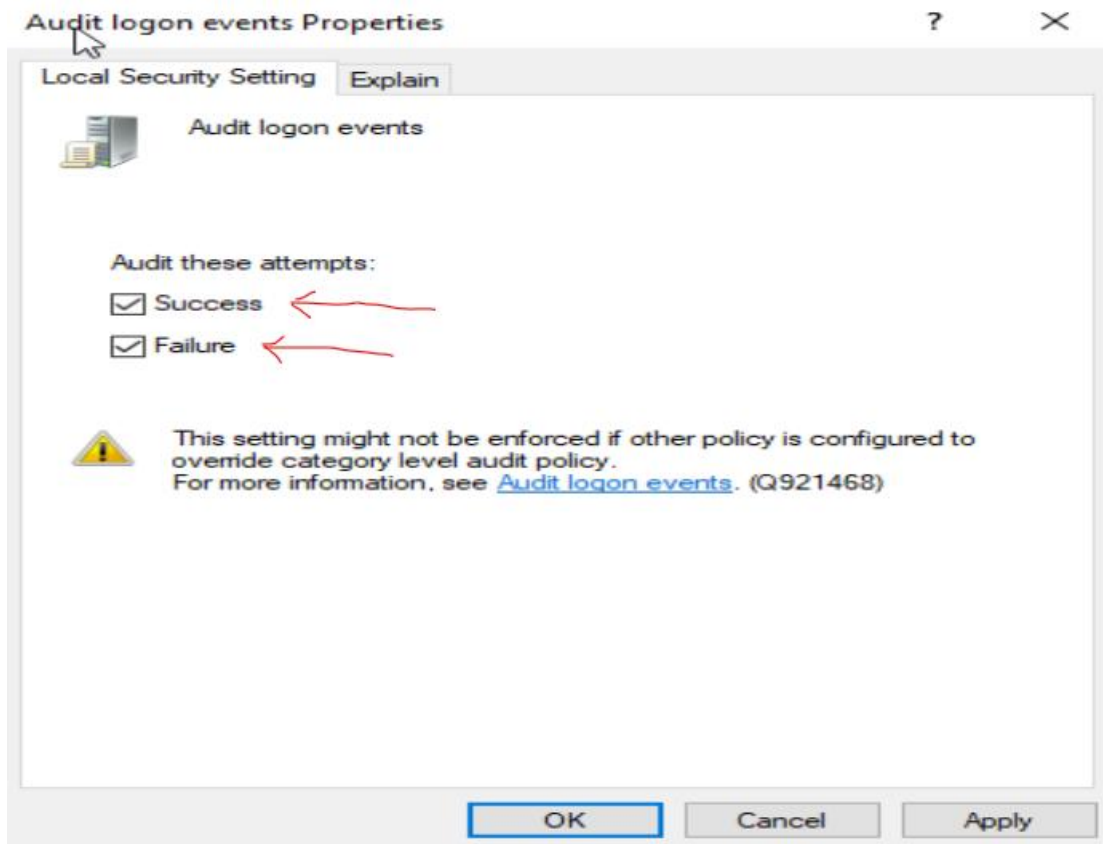
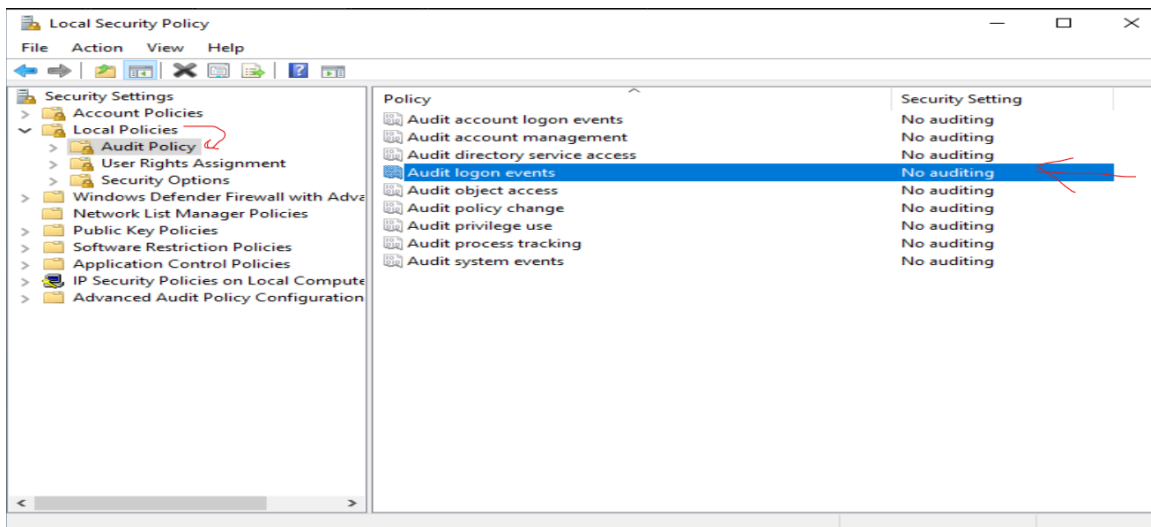


Filebeat and Winlogbeat are configured correctly.

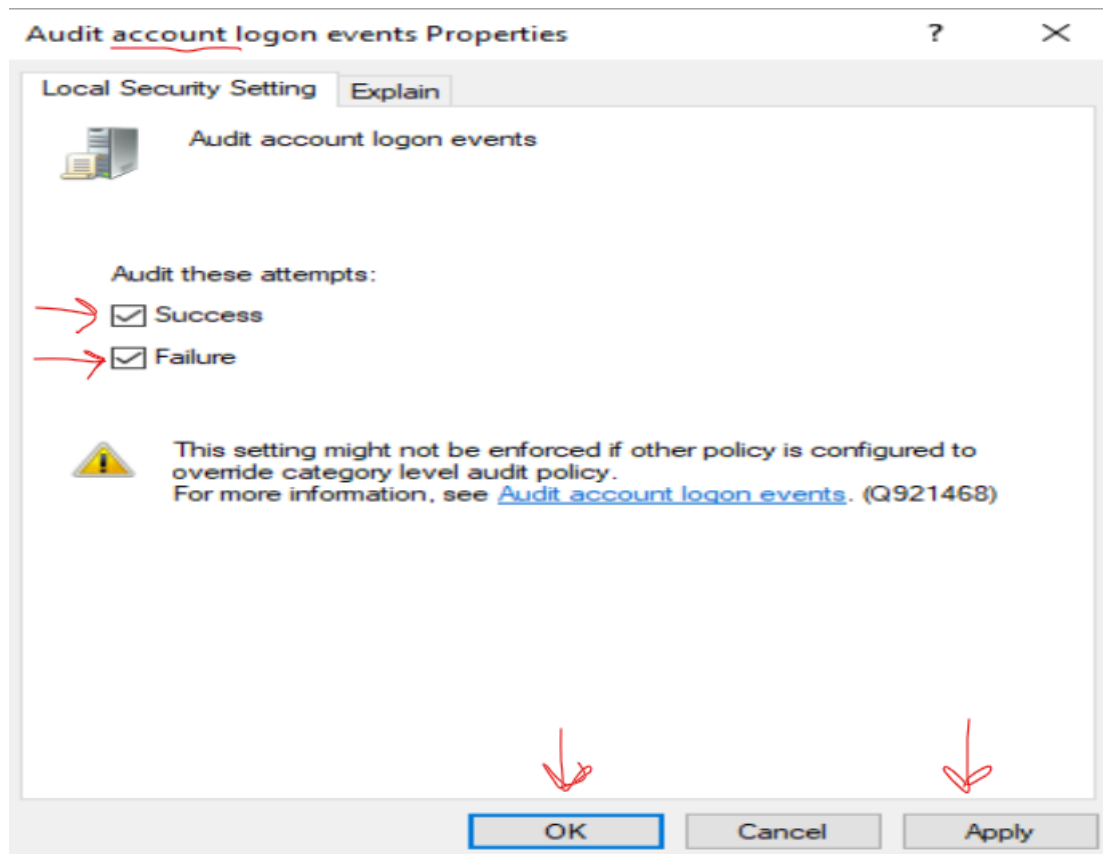
15. Enable the "Audit Logon Events" policy on Windows (otherwise failed logins — Event ID 4625 — won't be generated).



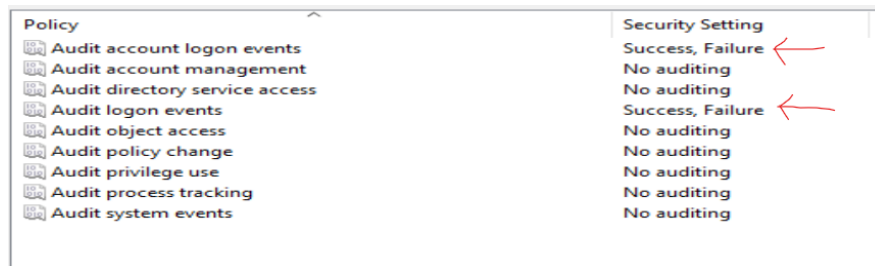
Local Security Policy → Local Policies → Audit Policy → Audit logon events



APPLY → OK



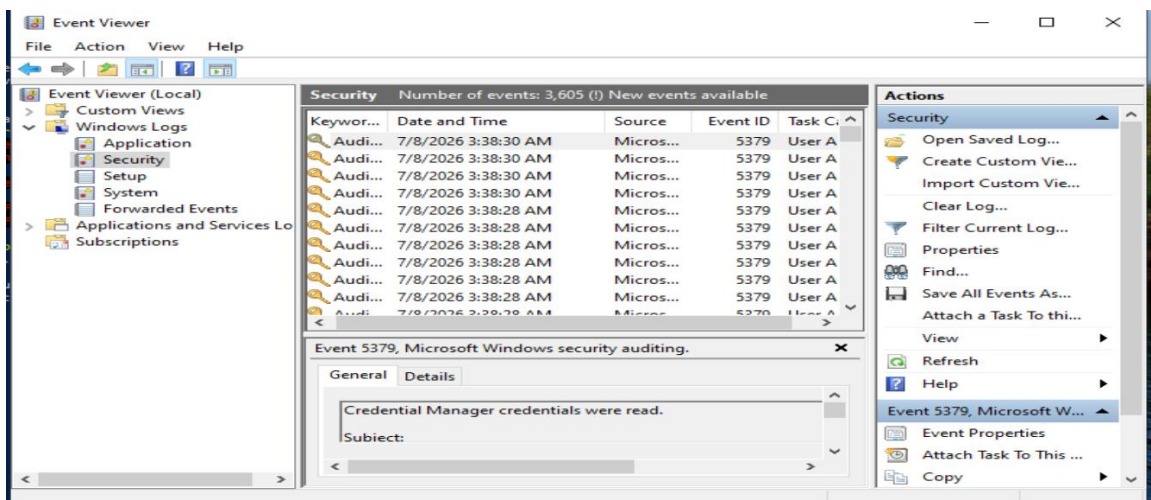
Audit account logon events — I also enable auditing of these attempts.



```
PS C:\Users\A> gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.
```

Audit Logon Events is enabled so that both successful and failed login attempts on Windows are recorded in the Security log and can later be analyzed in ELK/Kibana.



I filter to check whether Event ID 4625 appears. First I restart, enter wrong passwords, then check via Event Viewer.

Filter Current Log

Filter Current Log

Filter XML

Logged: Any time

Event level: ☐ Critical ☐ Warning ☐ Verbose ☐ Error ☐ Information

☒ By log Event logs: Security

☐ By source Event sources:

Includes/Excludes Event IDs: Enter ID numbers and/or ID ranges separated by commas. To exclude criteria, type a minus sign first. For example 1,3,5-99,-76

4625

Task category:

Keywords:

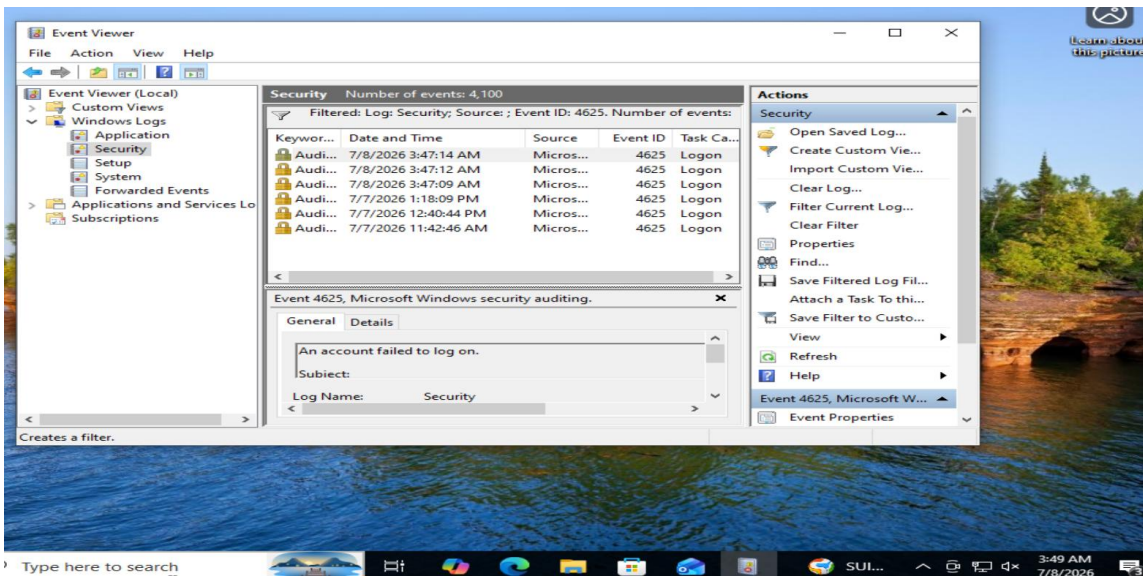
User: <All Users>

Computer(s): <All Computers>

Clear

OK Cancel

OK



The logs land here.

16. Install Winlogbeat or Elastic Agent on Windows and forward the logs to ELK.

```
PS C:\Program Files\Winlogbeat-Data\Winlogbeat> cd C:\Users\A\Downloads\winlogbeat-9.4.2-windows-x86_64\winlogbeat-9.4.2-windows-x86_64
PS C:\Users\A\Downloads\winlogbeat-9.4.2-windows-x86_64\winlogbeat-9.4.2-windows-x86_64> .\winlogbeat.exe test config -c "C:\Program Files\Winlogbeat-Data\Winlogbeat\winlogbeat.yml"
Exiting: no outputs are defined, please define one under the output section
PS C:\Users\A\Downloads\winlogbeat-9.4.2-windows-x86_64\winlogbeat-9.4.2-windows-x86_64> .\winlogbeat.exe test output -c "C:\Program Files\Winlogbeat-Data\Winlogbeat\winlogbeat.yml"
Error initializing output: output type undefined
PS C:\Users\A\Downloads\winlogbeat-9.4.2-windows-x86_64\winlogbeat-9.4.2-windows-x86_64> notepad "C:\Program Files\Winlogbeat-Data\Winlogbeat\winlogbeat.yml"
PS C:\Users\A\Downloads\winlogbeat-9.4.2-windows-x86_64\winlogbeat-9.4.2-windows-x86_64> .\winlogbeat.exe test config -c "C:\Program Files\Winlogbeat-Data\Winlogbeat\winlogbeat.yml"
Config OK
PS C:\Users\A\Downloads\winlogbeat-9.4.2-windows-x86_64\winlogbeat-9.4.2-windows-x86_64>
```



```
.winlogbeat - Notepad
File Edit Format View Help
bookmark: "<BookmarkList>\r\n <Bookmark Channel='Windows PowerShell' RecordId='764'
IsCurrent='true' />\r\n</BookmarkList>"

winlogbeat.event_logs:
- name: Application
- name: Security
- name: System

output.elasticsearch:
  hosts: ["http://192.168.10.22:9200"]

setup.kibana:
  host: "192.168.10.22:5601"

# These settings control loading the sample dashboards to the Kibana index. Loading
# the dashboards is disabled by default and can be enabled either by setting the
# options here or by using the `setup` command.
#setup.dashboards.enabled: false

# The URL from where to download the dashboard archive. By default, this URL
# has a value that is computed based on the Beat name and version. For released
# versions, this URL points to the dashboard archive on the artifacts.elastic.co
# website.
#setup.dashboards.url:

# ----- Kibana -----
# Starting with Beats version 6.0.0, the dashboards are loaded via the Kibana API.
# This requires a Kibana endpoint configuration.
setup.kibana:
  host: "http://192.168.10.22:5601"
  # Kibana Host
  # Scheme and port can be left out and will be set to the default (http and 5601)
  # In case you specify and additional path, the scheme is required: http://localhost:5601/
  # IPv6 addresses should always be defined as: https://[2001:db8::1]:5601
  #host: "localhost:5601"

# Kibana Space ID
```

I added these sections for configuration.

```
Config OK
C:\Program Files\winlogbeat> notepad "C:\Program Files\winlogbeat\winlogbeat.yml"
PS C:\Users\A\Downloads\winlogbeat-9.4.2-windows-x86_64\winlogbeat-9.4.2-windows-x86_64> Restart-Service winlogbeat
```

I also restart it so logs arrive as soon as ELK is open.

```
PS C:\Users\A\Downloads\winlogbeat-9.4.2-windows-x86_64\winlogbeat-9.4.2-windows-x86_64> Copy-Item "C:\Program Files\winlogbeat\winlogbeat.yml" "C:\Program Files\winlogbeat\winlogbeat.yml"
PS C:\Users\A\Downloads\winlogbeat-9.4.2-windows-x86_64\winlogbeat-9.4.2-windows-x86_64> .\winlogbeat.exe test config -c "C:\Program Files\winlogbeat\winlogbeat.yml"
Config OK
PS C:\Users\A\Downloads\winlogbeat-9.4.2-windows-x86_64\winlogbeat-9.4.2-windows-x86_64> .\winlogbeat.exe test output -c "C:\Program Files\winlogbeat\winlogbeat.yml"
elasticsearch: http://192.168.10.22:9200...
parse url... OK
connection... OK
parse host... OK
dns lookup... OK
addresses: 192.168.10.22
dial up... OK
talk to server... OK
version: 9.4.2
PS C:\Users\A\Downloads\winlogbeat-9.4.2-windows-x86_64\winlogbeat-9.4.2-windows-x86_64>
```

Talk to Server: OK

```
GNU nano 8.7.1 /etc/elasticsearch/elasticsearch.yml
#
# For more information, consult the discovery and cluster formation module documentation.
#
# ----- Various -----
#
# Allow wildcard deletion of indices:
#
#action.destructive_requires_name: false

#----- BEGIN SECURITY AUTO CONFIGURATION -----
#
# The following settings, TLS certificates, and keys have been automatically
# generated to configure Elasticsearch security features on 22-06-2026 15:57:39
#
# -----
#
# Enable security features
xpack.security.enabled: true

xpack.security.enrollment.enabled: true

# Enable encryption for HTTP API client connections, such as Kibana, Logstash, and Agents
xpack.security.http.ssl:
  enabled: true
  keystore.path: certs/http.p12

# Enable encryption and mutual authentication between cluster nodes
xpack.security.transport.ssl:
  enabled: true
  verification_mode: certificate
  keystore.path: certs/transport.p12
  truststore.path: certs/transport.p12
# Create a new cluster with the current node only
# Additional nodes can still join the cluster later
cluster.initial_master_nodes: ["elk1"]

# Allow HTTP API connections from anywhere
# Connections are encrypted and require user authentication
http.host: 0.0.0.0

# Allow other nodes to join the cluster from anywhere
# Connections are encrypted and mutually authenticated
#transport.host: 0.0.0.0

#----- END SECURITY AUTO CONFIGURATION -----
```

```
elk@elk:~$ sudo systemctl restart elasticsearch
elk@elk:~$ _
```

I reset the password.

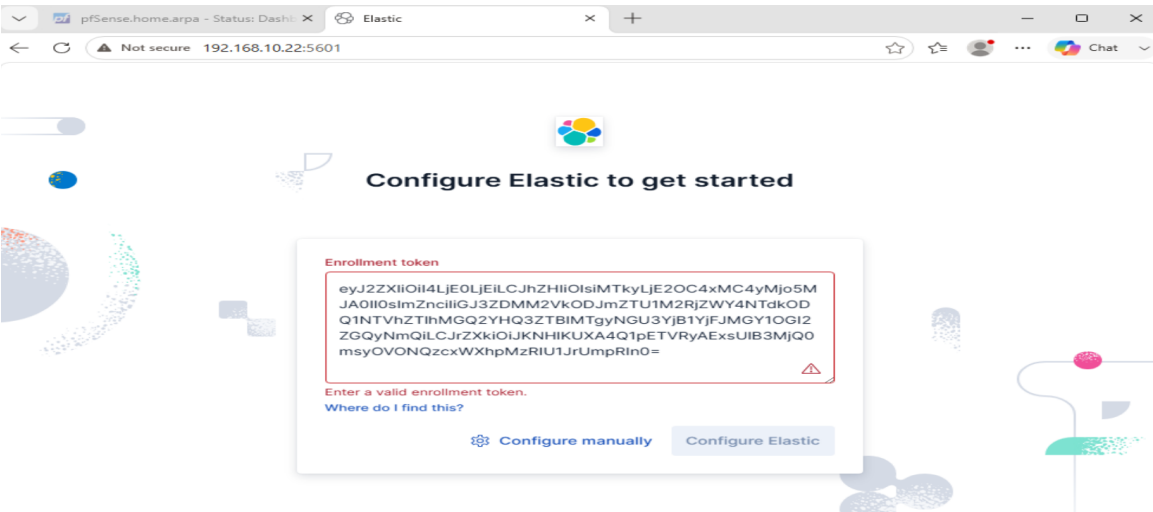
```

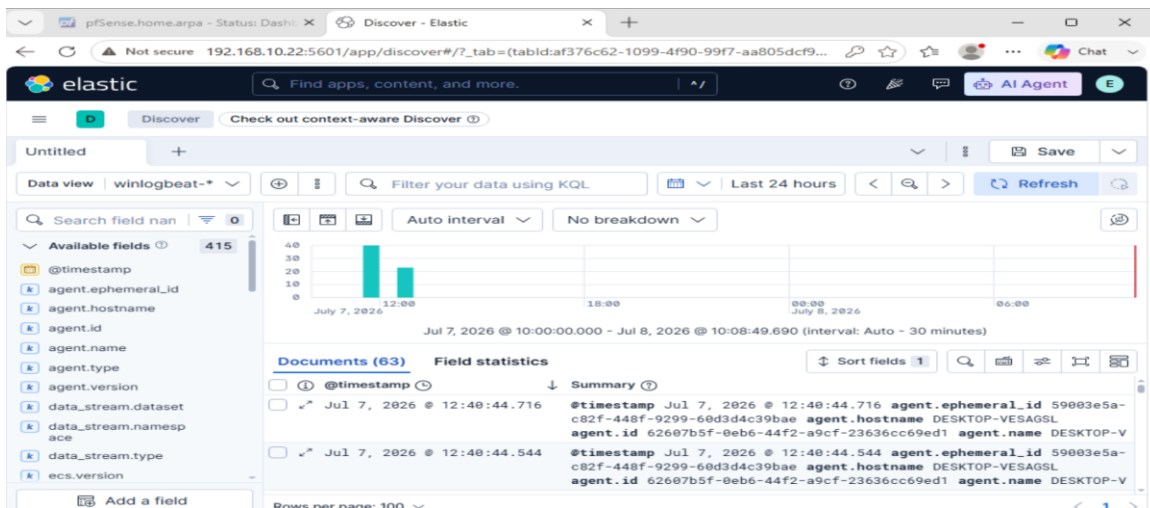
elk@elk:~$ sudo /usr/share/elasticsearch/bin/elasticsearch-reset-password -u elastic -i --url https://localhost:9200
This tool will reset the password of the [elastic] user.
You will be prompted to enter the password.
Please confirm that you would like to continue [y/N]y

Enter password for [elastic]:
Re-enter password for [elastic]:
Password for the [elastic] user successfully reset.
elk@elk:~$ sudo /usr/share/elasticsearch/bin/elasticsearch-reset-password -u kibana_system -i --url https://localhost:9200
This tool will reset the password of the [Kibana_system] user.
You will be prompted to enter the password.
Please confirm that you would like to continue [y/N]y

Enter password for [kibana_system]:
Re-enter password for [kibana_system]:
Password for the [kibana_system] user successfully reset.
elk@elk:~$ _

```





17. Integrate pfSense and Suricata logs into ELK (remote syslog; EVE JSON + Filebeat for Suricata).

```

elk@elk:/tmp$ sudo dpkg --configure -a
[sudo: authenticate] Password:
Setting up bpftool (7.7.0+7.0.0-27.27)...
dpkg: dependency problems prevent configuration of linux-main-modules-zfs-7.0.0-27-generic:
 linux-main-modules-zfs-7.0.0-27-generic depends on linux-image-7.0.0-27-generic | linux-image-unsigned-7.0.0-27-generic; however:
  Package linux-image-7.0.0-27-generic is not installed.
  Package linux-image-unsigned-7.0.0-27-generic is not installed.

dpkg: error processing package linux-main-modules-zfs-7.0.0-27-generic (--configure):
 dependency problems - leaving unconfigured
Processing triggers for man-db (2.13.1-1build1)...
Errors were encountered while processing:
 linux-main-modules-zfs-7.0.0-27-generic
elk@elk:/tmp$

```

It asked for a password, then returned a dependency error.

```

Use 'sudo apt autoremove' to remove them.

Upgrading:
 linux-modules-7.0.0-27-generic

Installing dependencies:
 linux-image-7.0.0-27-generic

Suggested packages:
 linux-tools
 linux-headers-7.0.0-27-generic

Summary:
Upgrading: 1, Installing: 1, Removing: 0, Not Upgrading: 49
2 not fully installed or removed.
Download size: 0.0 / 165 MB
Space needed: 184 MB / 7,094 MB available
 in /boot: 100 MB / 1,816 MB available

(Reading database, 270128 files and directories currently installed.)
Preparing to unpack ./linux-modules-7.0.0-27-generic-7.0.0-27.27.amd64.deb...
Unpacking linux-modules-7.0.0-27-generic (7.0.0-27.27) over (7.0.0-27.27)...
Selecting previously unselected package linux-image-7.0.0-27-generic.
Preparing to unpack ./linux-image-7.0.0-27-generic-7.0.0-27.27.amd64.deb...
Unpacking linux-image-7.0.0-27-generic (7.0.0-27.27)...
Setting up linux-modules-7.0.0-27-generic (7.0.0-27.27)...
Setting up linux-image-7.0.0-27-generic (7.0.0-27.27)...
I: /boot/vmlinuz is now a symlink to vmlinuz-7.0.0-27-generic
I: /boot/initrd.img is now a symlink to initrd.img-7.0.0-27-generic
Setting up linux-main-modules-zfs-7.0.0-27-generic (7.0.0-27.27)...
Processing triggers for linux-image-7.0.0-27-generic (7.0.0-27.27)...
/etc/kernel/postinst.d/dracut:
dracut: Generating /boot/initrd.img-7.0.0-27-generic
/etc/kernel/postinst.d/kdump-tools:
kdump-tools: Generating /var/lib/kdump/initrd.img-7.0.0-27-generic
/etc/kernel/postinst.d/zz-update-grub:
Sourcing file /etc/default/grub
Sourcing file /etc/default/grub.d/kdump-tools.cfg
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-7.0.0-27-generic
Found initrd image: /boot/initrd.img-7.0.0-27-generic
Found linux image: /boot/vmlinuz-7.0.0-27-generic
Found initrd image: /boot/vmlinuz-7.0.0-27-generic
Found initrd image: /boot/initrd.img-7.0.0-27-generic
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
needrestart is being skipped since dpkg has failed
elk@elk:/tmp$

```

Command → "sudo apt --fix-broken install -y" → used on Linux to automatically fix incompletely or improperly installed packages and pull in missing dependencies.

```

elk@elk:/tmp$ sudo dpkg --configure -a
elk@elk:/tmp$ sudo apt --fix-broken install -y
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 49
elk@elk:/tmp$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:4 https://artifacts.elastic.co/packages/9.x/apt stable InRelease
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [315 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [83.4 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [197 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [271 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [130 kB]
Fetched 1,271 kB in 2s (697 kB/s)
49 packages can be upgraded. Run 'apt list --upgradable' to see them.

```

I then run "sudo dpkg --configure -a" and "sudo apt --fix-broken install -y" again, followed by "sudo apt update".

I install Filebeat on ELK. To receive pfSense's remote syslog messages, a separate Filebeat instance also needs to be installed on the ELK server itself.

```

elk@elk:/tmp$ sudo apt install filebeat -y
Installing:
  filebeat

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 49
  Download size: 70.3 MB
  Space needed: 267 MB / 6,001 MB available

Get:1 https://artifacts.elastic.co/packages/9.x/apt stable/main amd64 filebeat amd64 9.4.3 [70.3 MB]
57% [1 filebeat 50.3 MB/70.3 MB 72%] 7,056 kB/s 2s

elk@elk:/tmp$ sudo systemctl status filebeat
* filebeat.service - Filebeat sends log files to Logstash or directly to Elasticsearch.
   Loaded: loaded (/usr/lib/systemd/system/filebeat.service; disabled; preset: enabled)
   Active: inactive (dead)
     Docs: https://www.elastic.co/beats/filebeat
elk@elk:/tmp$

```

It's inactive, so I start it.

```

elk@elk:/tmp$ sudo systemctl start filebeat
elk@elk:/tmp$ sudo systemctl status filebeat
* filebeat.service - Filebeat sends log files to Logstash or directly to Elasticsearch.
   Loaded: loaded (/usr/lib/systemd/system/filebeat.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-07-08 21:31:04 UTC; 1s ago
     Invocation: 502f4d04936945019470de0e4032f82a
       Docs: https://www.elastic.co/beats/filebeat
    Main PID: 14548 (filebeat)
      Tasks: 9 (limit: 1444)
     Memory: 198.2M (peak: 198.2M)
        CPU: 1.345s
    CGroup: /system.slice/filebeat.service
            └─14548 /usr/share/filebeat/bin/filebeat --environment systemd -c /etc/filebeat/filebeat.yml --path.home /usr/share/filebeat --path.config /etc/fi

Jul 08 21:31:05 elk filebeat[14548]: {"log.level":"info","@timestamp":"2026-07-08T21:31:05.538Z","log.logger":"modules","log.origin":{"function":"github.com/el
Jul 08 21:31:05 elk filebeat[14548]: {"log.level":"info","@timestamp":"2026-07-08T21:31:05.538Z","log.logger":"monitoring","log.origin":{"function":"github.com/el
Jul 08 21:31:05 elk filebeat[14548]: {"log.level":"info","@timestamp":"2026-07-08T21:31:05.538Z","log.logger":{"function":"github.com/elastic/beats/v7/libbeat/b
Jul 08 21:31:05 elk filebeat[14548]: {"log.level":"info","@timestamp":"2026-07-08T21:31:05.540Z","log.logger":"filebeat","log.origin":{"function":"github.com/el
Jul 08 21:31:05 elk filebeat[14548]: {"log.level":"info","@timestamp":"2026-07-08T21:31:05.540Z","log.logger":"registrar","log.origin":{"function":"github.com/el
Jul 08 21:31:05 elk filebeat[14548]: {"log.level":"info","@timestamp":"2026-07-08T21:31:05.540Z","log.logger":"crawler","log.origin":{"function":"github.com/el
Jul 08 21:31:05 elk filebeat[14548]: {"log.level":"info","@timestamp":"2026-07-08T21:31:05.540Z","log.logger":"crawler","log.origin":{"function":"github.com/el
Jul 08 21:31:05 elk filebeat[14548]: {"log.level":"info","@timestamp":"2026-07-08T21:31:05.541Z","log.logger":"crawler","log.origin":{"function":"github.com/el
Jul 08 21:31:05 elk filebeat[14548]: {"log.level":"info","@timestamp":"2026-07-08T21:31:05.541Z","log.logger":"crawler.module.reloader","log.origin":{"function":
Jul 08 21:31:05 elk filebeat[14548]: {"log.level":"info","@timestamp":"2026-07-08T21:31:05.541Z","log.logger":"crawler.module.reloader","log.origin":{"function":
lines 1-22/22 (END)

```

I made changes in the filebeat.yml file.

```

elk@elk:~$ sudo nano /etc/filebeat/filebeat.yml

```

```
# ===== Filebeat inputs =====
filebeat.inputs:
  - type: udp
    host: "0.0.0.0:5514"
    fields:
      log_type: pfsense_syslog
      fields_under_root: true

# Each - is an input. Most options can be set at the input level, so
# you can use different inputs for various configurations.
# Below are the input-specific configurations.

# filestream is an input for collecting log messages from files.
- type: filestream

# Unique ID among all inputs, an ID is required.
id: my-filestream-id

# Change to true to enable this input configuration.
enabled: false

# Paths that should be crawled and fetched. Glob based paths.
paths:
  - /var/log/*.log
  #- c:\programdata\elasticsearch\logs\*
```

```
# ----- Elasticsearch Output -----
output.elasticsearch:
  hosts: ["https://192.168.10.22:9200"]
  username: "elastic"
  password: "Salam123"
  ssl:
    verification_mode: none

# Array of hosts to connect to.
hosts: ["localhost:9200"]

# Performance preset - one of "balanced", "throughput", "scale",
# "latency", or "custom".
preset: balanced

# Protocol - either `http` (default) or `https`.
#protocol: "https"

# Authentication credentials - either API key or username/password.
#api_key: "id:api_key"
#username: "elastic"
#password: "changeme"
```

```
# ===== Kibana =====

# Starting with Beats version 6.0.0, the dashboards are loaded via the Kibana API.
# This requires a Kibana endpoint configuration.
setup.kibana:
  host: "http://192.168.10.22:5601"
  # Kibana Host
  # Scheme and port can be left out and will be set to the default (http and 5601)
  # In case you specify an additional path, the scheme is required: http://localhost:5601/path
  # IPv6 addresses should always be defined as: https://[2001:db8::1]:5601
  #host: "localhost:5601"

# Kibana Space ID
# ID of the Kibana Space into which the dashboards should be loaded. By default,
# the Default Space will be used.
#space.id:
```

CTRL+O → ENTER → CTRL+X

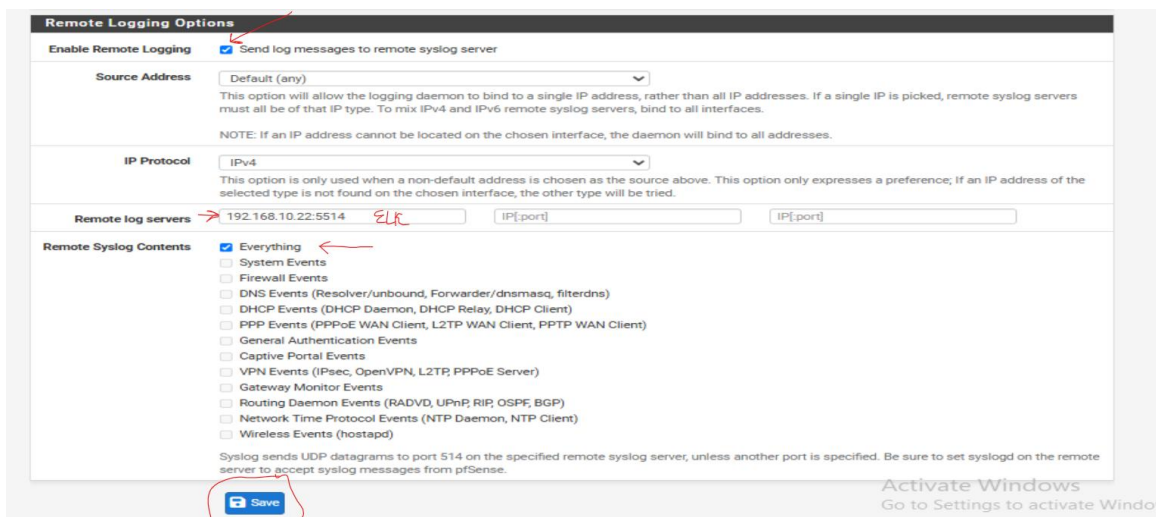
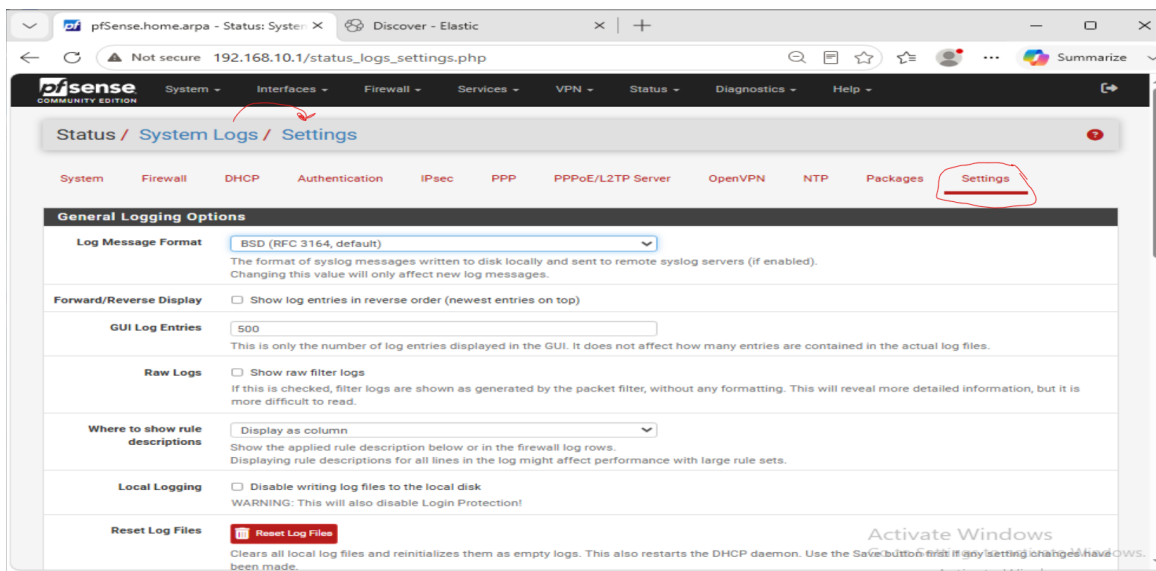
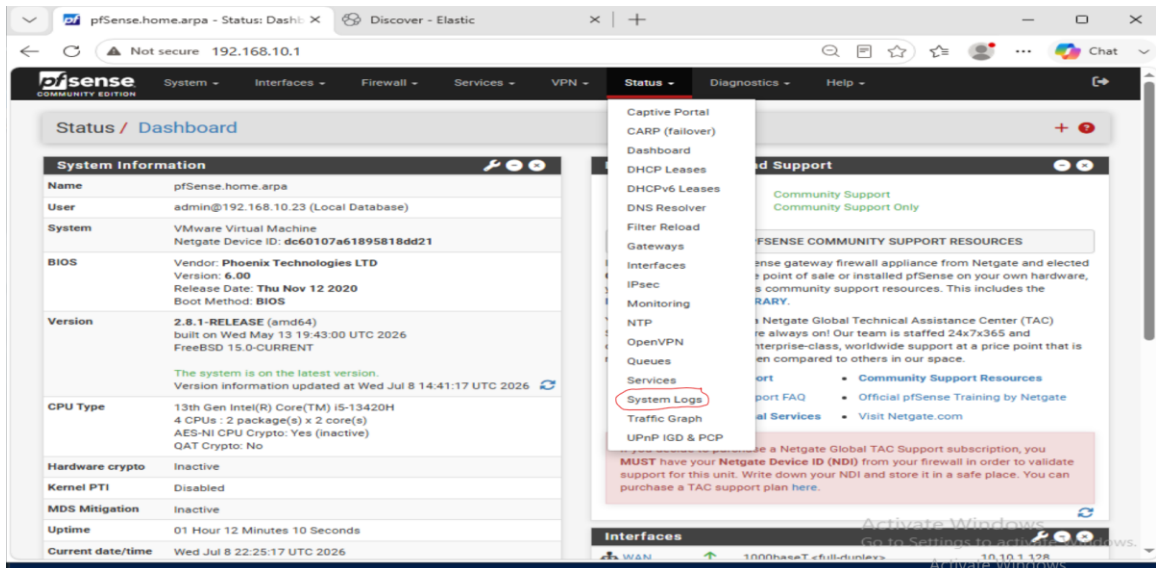
```
elk@elk:~$ sudo filebeat test config
Config OK
```

```
elk@elk:~$ sudo filebeat test output
elasticsearch: https://192.168.10.22:9200...
  parse url... OK
  connection...
    parse host... OK
    dns lookup... OK
    addresses: 192.168.10.22
    dial up... OK
  TLS...
    security... WARN server's certificate chain verification is disabled
    handshake... OK
    TLS version: TLSv1.3
    dial up... OK
  talk to server... OK ←
  version: 9.4.2
elk@elk:~$
```

I restart Filebeat.

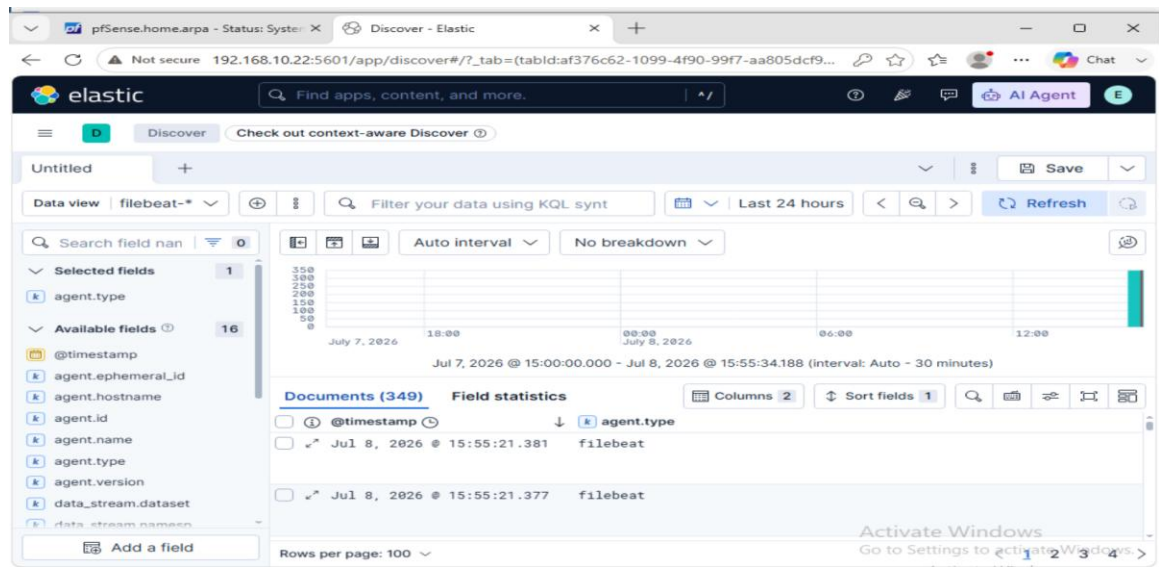
```
elk@elk:~$ sudo systemctl restart filebeat
elk@elk:~$ sudo ss -ulnp | grep 5514
UNCONN 0      0      *:5514      *:5514      users:((("filebeat",pid=2697,fd=9))
elk@elk:~$
```

Now I switch to pfSense.



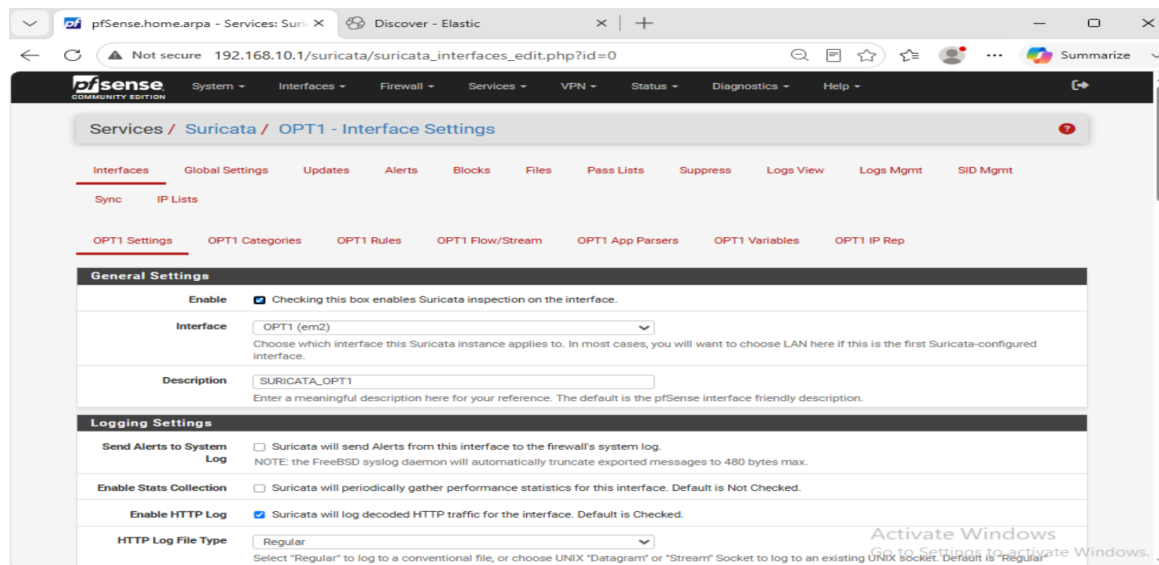
SAVE

If needed: Stack Management → Kibana: Data View → New Data View → Create

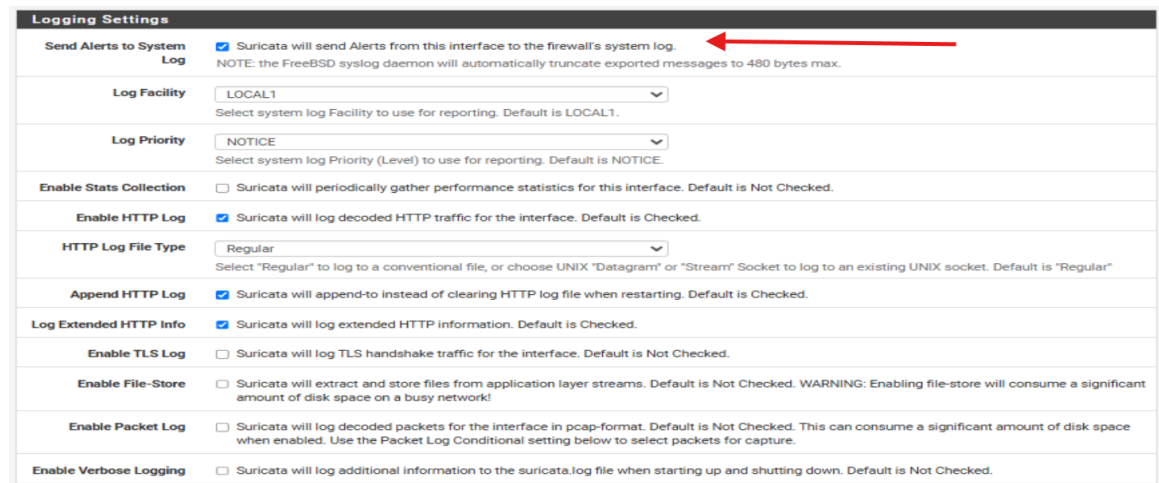


pfSense log integration into ELK is complete.

Now switching to SURICATA.



Logging Settings



EVE Output Settings

EVE JSON Log ☒ Suricata will output selected info in JSON format to a single file or to syslog. Default is Not Checked. 

EVE Output Type FILE
Select EVE log output destination. Choosing FILE is suggested and is the default value. "Redis" is used for output to a Redis server, and the UNIX Socket options output to a user-created socket.

EVE HTTP XFF Support ☐ Log X-Forwarded-For IP addresses. Default is Not Checked.

EVE Ethernet MAC ☐ Log Ethernet header in events when available. Default is Not Checked.

EVE Log Alerts ☒ Suricata will output Alerts via EVE

EVE Log Alert Payload Data Formats BOTH
Log the payload data with alerts. Options are No (disable payload logging), Only Printable (lossy) format, Only Base64 encoded or Both. See Suricata documentation.

EVE Log Alert details ☒ Log a packet dump with alerts. ☒ Log additional HTTP data. ☒ Include App Layer metadata. ☐ Log final action taken on packet by the engine. ☐ Log packets for rules using the "tag" keyword

EVE Log Drops ☒ Suricata will output Drops via EVE

EVE Log Drops Options ☒ Log alerts that caused drops. Default is "Checked". ☐ Log final action taken on packet by the engine. All
"Start" logs only a single drop per flow direction. "All" logs each dropped pkt.

EVE Log Anomalies ☐ Suricata will log packet anomalies such as truncated packets, packets with invalid IP/UDP/TCP length values and other events that render the packet invalid for further processing. Networks with high rates of anomalies may experience packet processing degradation.

EVE Logged Traffic ☐ BitTorrent ☒ DNS ☒ FTP ☒ HTTP ☒ HTTP2 ☒ IKE ☒ Kerberos ☒ NFS ☐ PostgreSQL ☒ QUIDV1 ☐ RDP ☒ RFB ☐ SIP ☒ SMB ☒ SMTP ☒ TFTP

SAVE — then I restart it.

EXAMPLE

kali@kali: ~/Desktop

Session Actions Edit View Help

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~[~/Desktop]
$ ping 192.168.10.23
PING 192.168.10.23 (192.168.10.23) 56(84) bytes of data.
64 bytes from 192.168.10.23: icmp_seq=1 ttl=127 time=8.99 ms
64 bytes from 192.168.10.23: icmp_seq=2 ttl=127 time=2.37 ms
64 bytes from 192.168.10.23: icmp_seq=3 ttl=127 time=2.38 ms
64 bytes from 192.168.10.23: icmp_seq=4 ttl=127 time=2.97 ms
64 bytes from 192.168.10.23: icmp_seq=5 ttl=127 time=2.81 ms
64 bytes from 192.168.10.23: icmp_seq=6 ttl=127 time=2.44 ms
64 bytes from 192.168.10.23: icmp_seq=7 ttl=127 time=2.04 ms
64 bytes from 192.168.10.23: icmp_seq=8 ttl=127 time=2.02 ms
64 bytes from 192.168.10.23: icmp_seq=9 ttl=127 time=2.89 ms
64 bytes from 192.168.10.23: icmp_seq=10 ttl=127 time=2.08 ms
64 bytes from 192.168.10.23: icmp_seq=11 ttl=127 time=2.52 ms
64 bytes from 192.168.10.23: icmp_seq=12 ttl=127 time=2.30 ms
64 bytes from 192.168.10.23: icmp_seq=13 ttl=127 time=2.13 ms
64 bytes from 192.168.10.23: icmp_seq=14 ttl=127 time=2.72 ms
64 bytes from 192.168.10.23: icmp_seq=15 ttl=127 time=1.70 ms
64 bytes from 192.168.10.23: icmp_seq=16 ttl=127 time=2.09 ms
64 bytes from 192.168.10.23: icmp_seq=17 ttl=127 time=2.04 ms
```

pfSense.home.arpa - Services: Suricata | Discover - Elastic

Not secure 192.168.10.1/suricata/suricata_alerts.php?instance=0

Services / Suricata / Alerts

Interfaces Global Settings Updates Alerts Blocks Files Pass Lists Suppress Logs View Logs Mgmt SID Mgmt

Sync IP Lists

Alert Log View Settings

Instance to View (OPT1) SURICATA_OPT1
Choose which instance alerts you want to inspect.

Save or Remove Logs Download Clear
All alert log files for selected interface will be downloaded. Clear the currently active Alerts log file

Save Settings Save ☒ Refresh 250
Save auto-refresh and view settings. Default is ON. Number of alerts to display: Default is 250

Alert Log View Filter

Last 250 Alert Entries. (Most recent entries are listed first)

Note: Alerts triggered by DROP rules that resulted in dropped (blocked) packets are shown with highlighted rows below.

Date	Action	Pri	Proto	Class	Src	SPort	Dest	DPort	GID:SID	Description
07/08/2026 23:05:21		3	ICMP	Not Assigned	192.168.20.20	8	192.168.10.23	0	1:1000001	PING FROM KALI TO WINDOWS 
07/07/2026 21:03:57		3	TCP	Not Assigned	192.168.20.20	43032	192.168.10.23	5061	1:1000002	CUSTOM Possible Nmap TCP Scan from Kali
07/07/2026 21:03:47		3	TCP	Not Assigned	192.168.20.20	43030	192.168.10.23	5963	1:1000002	CUSTOM Possible Nmap TCP Scan from Kali

The screenshot shows the Elastic Discover interface. On the left, the 'Selected fields' list includes @timestamp, agent.ephemeral_id, agent.hostname, agent.id, agent.name, agent.type, agent.version, data_stream.dataset, and data_stream.namespace. The 'Available fields' list includes @timestamp, agent.ephemeral_id, agent.hostname, agent.id, agent.name, agent.type, agent.version, data_stream.dataset, and data_stream.namespace. The main view shows a table of documents. The first document is selected, showing its fields and values:

Field	Value
agent.version	9.4.3
ecs.version	8.0.0
host.name	elk
input.type	udp
log_type	pfsense_syslog
log.source.address	192.168.10.1:514
message	<141>Jul 8 23:05:21 suricata[6178]: [1:1000001:1] PING FROM KALI TO WINDOWS (Classification: (nu...)) [Priority: 3] [ICMP] 192.168.20.20 -> 192.168.10.23:0

The log arrived.

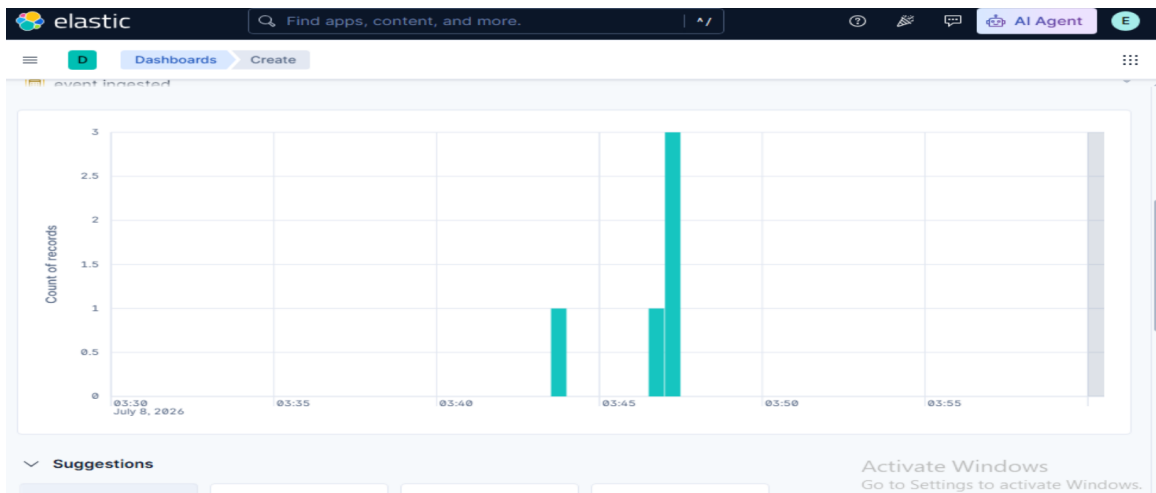
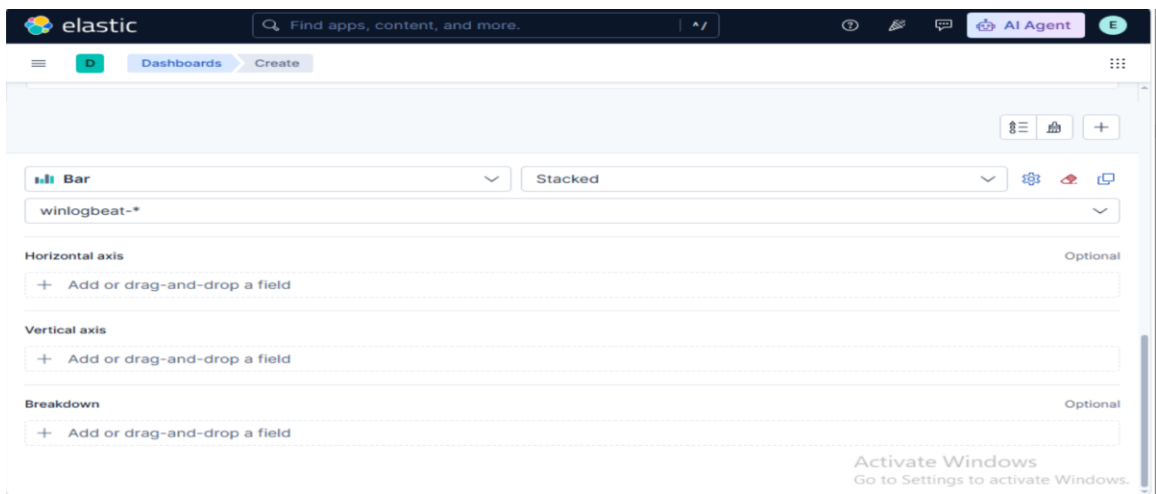
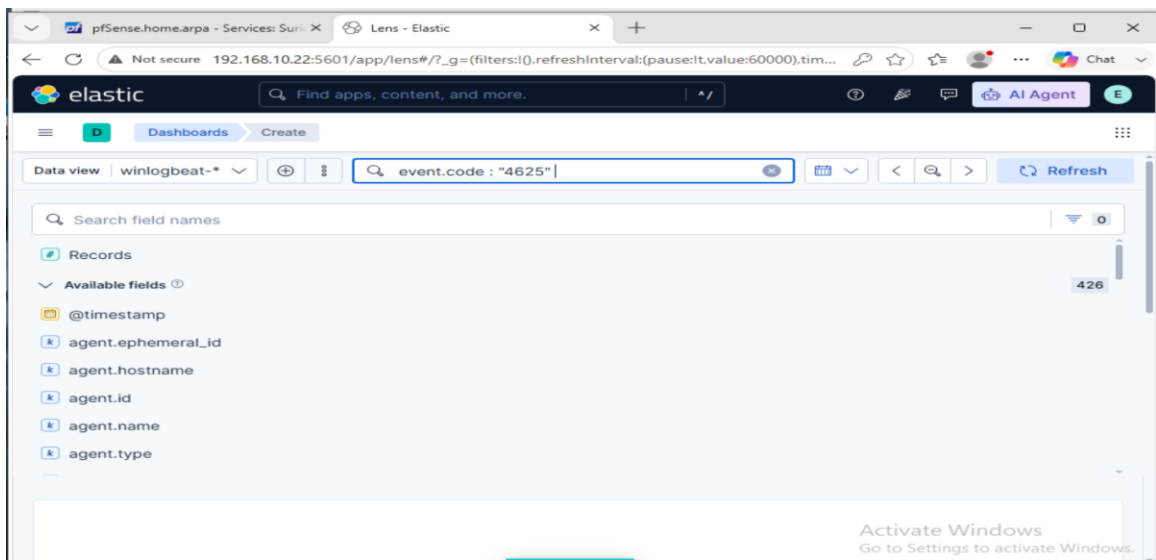
18. Build at least one Kibana dashboard: failed login attempts, Suricata alerts, firewall blocks.

The screenshot shows the Elastic Dashboards interface. The 'Dashboards' tab is selected. A red arrow points to the 'Create dashboard' button. The list of dashboards includes:

- [Winlogbeat Security] User Management Events - Simple Metric
- [Winlogbeat Security] User Management Events
- [Winlogbeat Security] User Logons - Simple Metrics
- [Winlogbeat Security] Group Management Events - Simple Metrics
- [Winlogbeat Security] Failed and Blocked Accounts - Simple Metrics

The screenshot shows the Elastic Dashboards interface with the 'Dashboard settings' dialog open. The 'Title' field is set to 'Project Monitoring Dashboard'. The 'Description' field is empty. The 'Tags' field is empty. The 'Store time with dashboard' checkbox is checked. The 'Apply' button is highlighted.

Create Visualization → Data View: winlogbeat-*



A bar chart appears here. SAVE AND RETURN

Firewall example for blocks as well.

```
(kali@kali)-[~/Desktop]
$ nmap -p 3389 192.168.10.23
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-08 19:58 -0400
Nmap scan report for 192.168.10.23
Host is up (0.032s latency).

PORT      STATE      SERVICE
3389/tcp  filtered  ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 1.13 seconds

(kali@kali)-[~/Desktop]
$
```

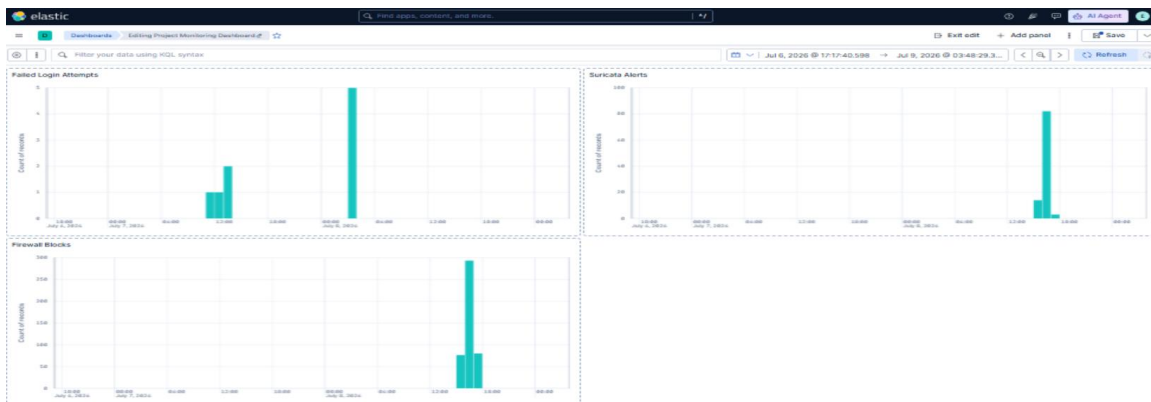
pfSense.home.arpa - Status: System | Project Monitoring Dashboard - ELK

Not secure 192.168.10.1/status_logs_filter.php

✗	Jul 8 23:58:15	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137	10.10.1.255:137	UDP
✓	Jul 8 23:58:15	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.22:33798	34.120.127.130:443	TCP-S
✗	Jul 8 23:58:15	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137	10.10.1.255:137	UDP
✗	Jul 8 23:58:22	LAN	Default deny rule IPv4 (1000000103)	192.168.10.23:57703	217.25.25.43:443	TCP-A
✗	Jul 8 23:58:24	WAN	Default deny rule IPv4 (1000000103)	98.66.133.184:443	10.10.1.128:7100	TCP-PA
✗	Jul 8 23:58:25	WAN	Default deny rule IPv4 (1000000103)	18.165.72.47:443	10.10.1.128:63884	TCP-FPA
✗	Jul 8 23:58:25	LAN	Default deny rule IPv4 (1000000103)	192.168.10.23:57686	150.171.27.11:443	TCP-A
✗	Jul 8 23:58:33	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137	10.10.1.255:137	UDP
✗	Jul 8 23:58:34	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137	10.10.1.255:137	UDP
✗	Jul 8 23:58:34	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137	10.10.1.255:137	UDP
✓	Jul 8 23:58:35	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.23:65005	10.10.1.131:1514	TCP-S
✗	Jul 8 23:58:36	LAN	Default deny rule IPv4 (1000000103)	192.168.10.23:57704	13.107.213.63:443	TCP-A
✗	Jul 8 23:58:36	LAN	Default deny rule IPv4 (1000000103)	192.168.10.23:57707	18.165.72.47:443	TCP-A
✗	Jul 8 23:58:40	OPT1	Default deny rule IPv4 (1000000103)	192.168.20.20:61754	192.168.10.23:80	TCP-A
✓	Jul 8 23:58:40	OPT1	ALLOW: KALI(OPT1) → Windows(LAN) for Pentest (1783378859)	192.168.20.20:61754	192.168.10.23:443	TCP-S
✓	Jul 8 23:58:40	OPT1	ALLOW: KALI(OPT1) → Windows(LAN) for Pentest (1783378859)	192.168.20.20	192.168.10.23	ICMP
✓	Jul 8 23:58:40	OPT1	ALLOW: KALI(OPT1) → Windows(LAN) for Pentest (1783378859)	192.168.20.20	192.168.10.23	ICMP
✗	Jul 8 23:58:41	OPT1	BLOCK KALI TO WINDOWS VIA RDP (1783379485)	192.168.20.20:62010	192.168.10.23:3389	TCP-S
✗	Jul 8 23:58:41	OPT1	BLOCK KALI TO WINDOWS VIA RDP (1783379485)	192.168.20.20:62012	192.168.10.23:3389	TCP-S
✗	Jul 8 23:58:44	LAN	Default deny rule IPv4 (1000000103)	192.168.10.23:57490	34.120.127.120:443	TCP-A
✓	Jul 8 23:58:44	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.22:35142	34.95.113.255:443	TCP-S

Activate Windows

The event arrived at pfSense as a log (BLOCK).



The dashboard is ready, with three separate panels.

Phase 5 — Attack and Detection (Kali)

Port Scan

Kali runs an nmap scan against Windows → pfSense sees the traffic → Suricata generates an alert for the scan → ELK/Kibana displays that alert

Kali: 192.168.20.20

Windows: 192.168.10.23

ELK: 192.168.10.22

```
(kali@kali)-[~/Desktop]
$ nmap -sS 192.168.10.23
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-08 20:26 -0400
Nmap scan report for 192.168.10.23
Host is up (0.019s latency).
All 1000 scanned ports on 192.168.10.23 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 21.94 seconds
```

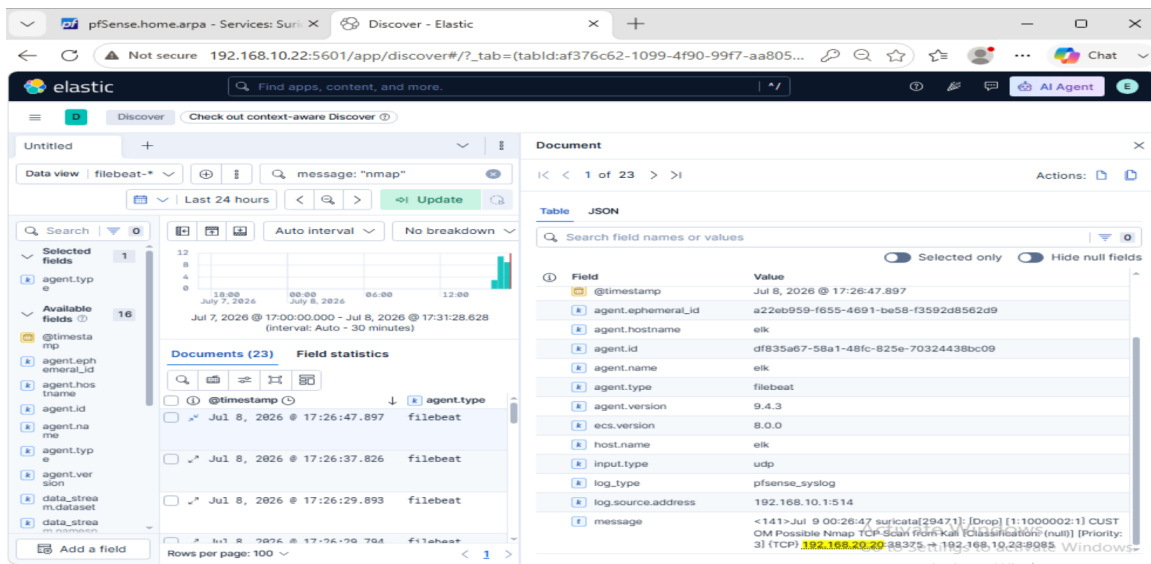
Kali runs an nmap scan against Windows

Time	Interface	Event Type	Details
Jul 9 00:27:00	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.23:65150 → 217.25.25.43:443 TCP-S
Jul 9 00:27:30	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.23:65153 → 10.10.1.131:1514 TCP-S
Jul 9 00:28:01	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.23:65154 → 10.10.1.131:1514 TCP-S
Jul 9 00:28:03	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137 → 10.10.1.255:137 UDP
Jul 9 00:28:04	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137 → 10.10.1.255:137 UDP
Jul 9 00:28:04	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137 → 10.10.1.255:137 UDP
Jul 9 00:28:05	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137 → 10.10.1.255:137 UDP
Jul 9 00:28:05	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137 → 10.10.1.255:137 UDP
Jul 9 00:28:06	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.22:60185 → 8.8.8.8:53 UDP
Jul 9 00:28:06	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.22:49587 → 8.8.8.8:53 UDP
Jul 9 00:28:06	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.22:53090 → 34.120.127.130:443 TCP-S
Jul 9 00:28:06	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137 → 10.10.1.255:137 UDP
Jul 9 00:28:07	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137 → 10.10.1.255:137 UDP
Jul 9 00:28:07	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137 → 10.10.1.255:137 UDP
Jul 9 00:28:08	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.22:45270 → 34.95.113.255:443 TCP-S
Jul 9 00:28:09	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137 → 10.10.1.255:137 UDP
Jul 9 00:28:09	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.23:57903 → 8.8.8.8:53 UDP
Jul 9 00:28:09	LAN	LAN-dan WAN-a internet cixisina icaze (1783378570)	192.168.10.23:50473 → 8.8.8.8:53 UDP
Jul 9 00:28:09	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137 → 10.10.1.255:137 UDP
Jul 9 00:28:10	WAN	Block private networks from WAN block 10/8 (12011)	10.10.1.1:137 → 10.10.1.255:137 UDP

pfSense sees the traffic

Date	Action	Pri	Proto	Class	Src	SPort	Dst	DPort	GID:SID	Description
07/09/2026 00:26:47	Alert	3	TCP	Not Assigned	192.168.20.20	38375	192.168.10.23	8085	1-1000002	CUSTOM Possible Nmap TCP Scan from Kali
07/09/2026 00:26:37	Alert	3	TCP	Not Assigned	192.168.20.20	38375	192.168.10.23	880	1-1000002	CUSTOM Possible Nmap TCP Scan from Kali
07/09/2026 00:26:29	Alert	3	TCP	Not Assigned	192.168.20.20	38377	192.168.10.23	3389	1-1000003	CUSTOM RDP Connection Attempt from Kali to Windows
07/09/2026 00:26:29	Alert	3	TCP	Not Assigned	192.168.20.20	38375	192.168.10.23	3389	1-1000003	CUSTOM RDP Connection Attempt from Kali to Windows
07/09/2026 00:26:28	Alert	3	TCP	Not Assigned	192.168.20.20	38375	192.168.10.23	21	1-1000002	CUSTOM Possible Nmap TCP Scan from Kali
07/09/2026 00:26:27	Alert	3	ICMP	Not Assigned	192.168.20.20	13	192.168.10.23	0	1-1000001	PING FROM KALI TO WINDOWS
07/09/2026 00:26:27	Alert	3	ICMP	Not Assigned	192.168.20.20	8	192.168.10.23	0	1-1000001	PING FROM KALI TO WINDOWS
07/08/2026	Alert	3	TCP	Not Assigned	192.168.20.20	62012	192.168.10.23	3389	1-1000003	CUSTOM RDP Connection Attempt from Kali to Windows

Suricata generates an alert for the scan



ELK/Kibana displays that alert

Brute-Force Login Attempt (e.g., Hydra against RDP/SMB) — should be visible in Windows logs and Kibana

Kali attempts RDP/SMB logins against Windows → Windows logs it as a failed login in the Security log → Winlogbeat forwards those Event ID 4625 events to ELK → visible in Kibana

Kali: 192.168.20.20

Windows: 192.168.10.23

Kibana: winlogbeat-*

```
kali@kali: ~/Desktop
Session Actions Edit View Help

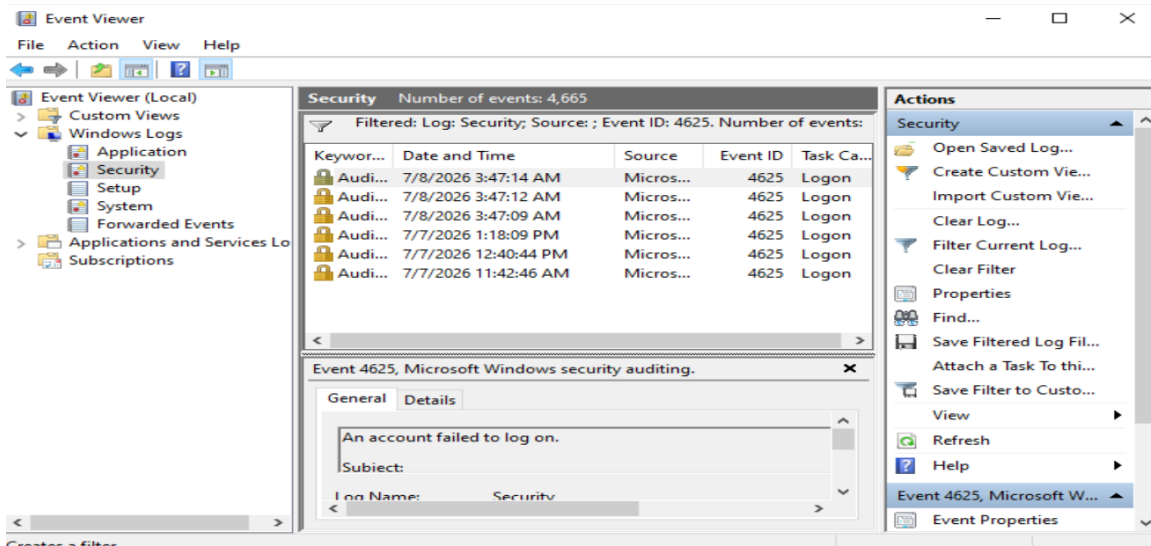
(kali@kali)~[~/Desktop]
$ nano imtahan-layihe-brute-force.txt

(kali@kali)~[~/Desktop]
$ cat imtahan-layihe-brute-force.txt
Salam123
kali
Ne_var_ne_yox
Cavab123
ubuntu0987

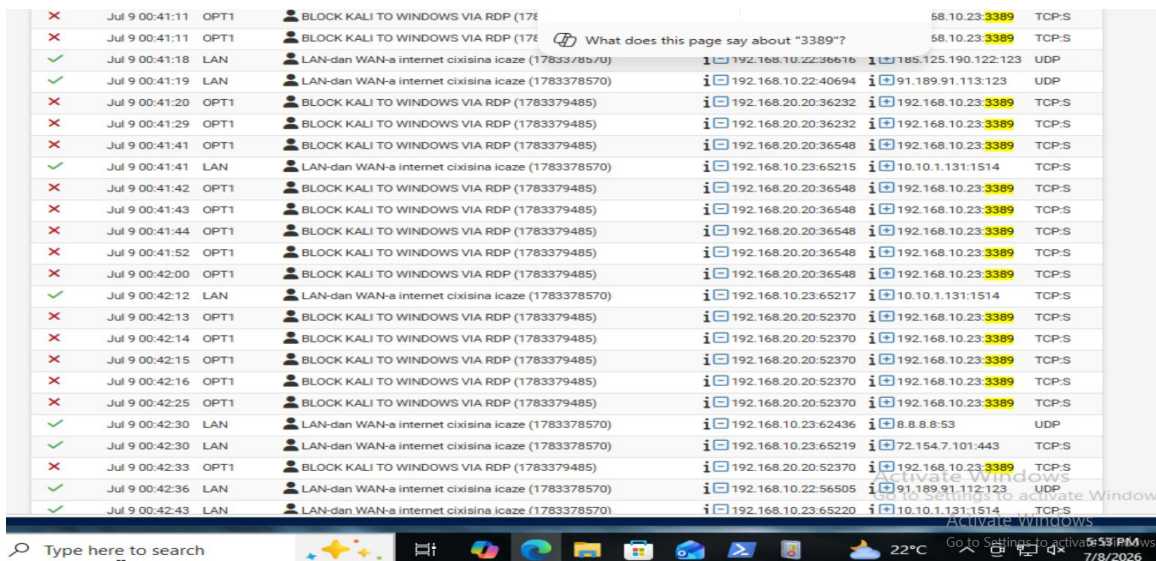
(kali@kali)~[~/Desktop]
$ hydra -l A -P imtahan-layihe-brute-force.txt rdp://192.168.10.23 -V -t 1
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-08 20:41:07
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 1 task per 1 server, overall 1 task, 5 login tries (l:1/p:5), ~5 tries per task
[DATA] attacking rdp://192.168.10.23:3389/
[ATTEMPT] target 192.168.10.23 - login "A" - pass "Salam123" - 1 of 5 [child 0] (0/0)
[ERROR] freerdp: The connection failed to establish.
[ATTEMPT] target 192.168.10.23 - login "A" - pass "kali" - 2 of 6 [child 0] (0/1)
[STATUS] 2.00 tries/min, 2 tries in 00:01h, 4 to do in 00:03h, 1 active
[ERROR] freerdp: The connection failed to establish.
[ATTEMPT] target 192.168.10.23 - login "A" - pass "Ne_var_ne_yox" - 3 of 7 [child 0] (0/2)
[ERROR] freerdp: The connection failed to establish.
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-08 20:42:44
```

Kali attempts to log into Windows RDP. Every wrong password should appear in the Windows Security log as a 4625 failed login event. Hydra tried all 5 passwords.



Windows Event Viewer recorded it. I used Filter Current Log and filtered on Event ID 4625 to view the list directly — in short, the logs are written to Event Viewer.



pfSense sees the traffic

In addition, to demonstrate the difference between IDS and IPS, I ran the network mapping scan from Kali again.

```
(kali@kali)-[~/Desktop]
$ nmap -p 3389 192.168.10.23
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-08 21:00 -0400
Nmap scan report for 192.168.10.23
Host is up (0.044s latency).

PORT      STATE      SERVICE
3389/tcp  filtered  ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 1.21 seconds
```

The "filtered" result here proves IPS is working in inline mode. "Filtered" means Kali tried to connect to the RDP port, but no normal response came back — meaning the traffic was blocked or dropped somewhere by the firewall/IPS.

Document

1 of 18

Actions: [Icons]

Table JSON

Search field names or values

Selected only Hide null fields

Field	Value
agent.ephemeral_id	a22eb959-f655-4691-be58-f3592d8562d9
agent.hostname	elk
agent.id	df835a67-58a1-48fc-825e-70324438bc09
agent.name	elk
agent.type	filebeat
agent.version	9.4.3
ecs.version	<141>Jul 9 01:00:01 suricata[29471]: [1:1000003:1] CUST
host.name	OM RDP Connection Attempt from Kali to Windows [Classi
input.type	fication: (null)] [Priority: 3] {TCP} 192.168.20.20:52267 →
log_type	192.168.10.23:3389
log.source	

Filter for value Filter out value Copy value

Network

Kali: 192.168.20.20

Windows: 192.168.10.23

ELK: 192.168.10.22

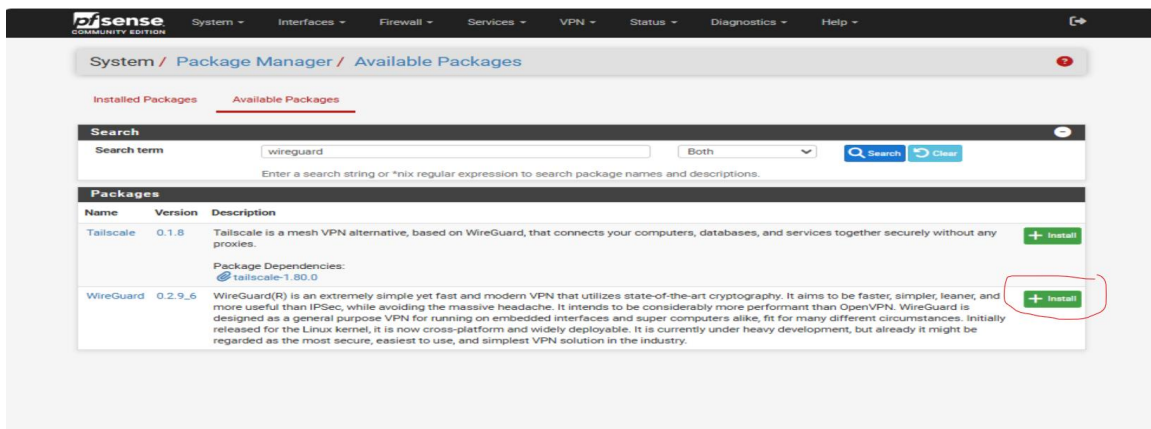
pfSense LAN: 192.168.10.1

pfSense OPT1: 192.168.20.1

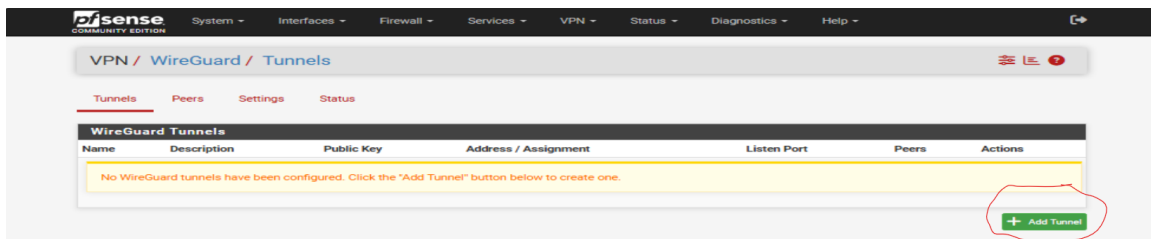
pfSense WAN: 10.10.1.128/24

Bonus Task

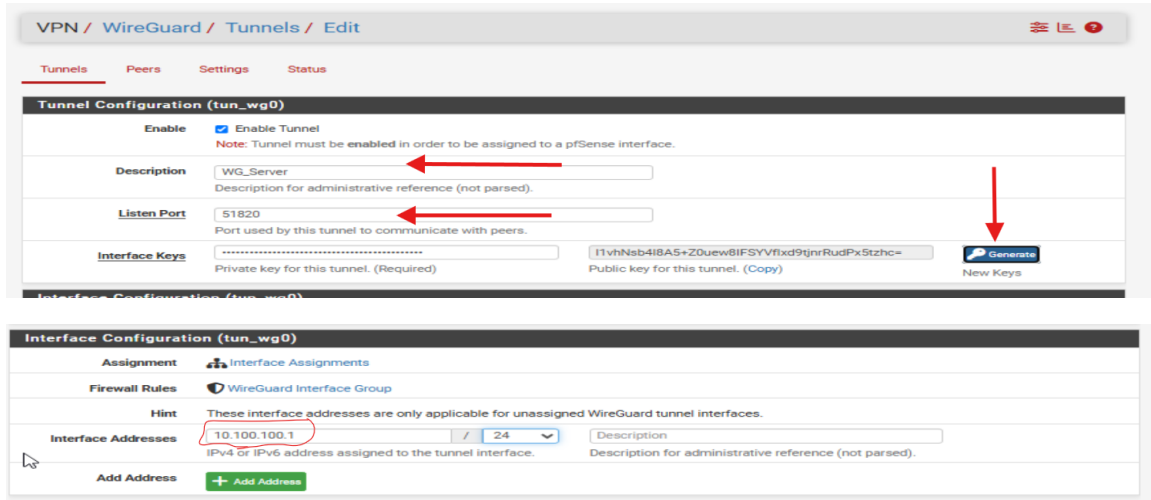
Add VPN configuration to pfSense (OpenVPN/WireGuard)



I first installed the package, then went to the VPN section: WireGuard → Tunnels → Add Tunnel (select)

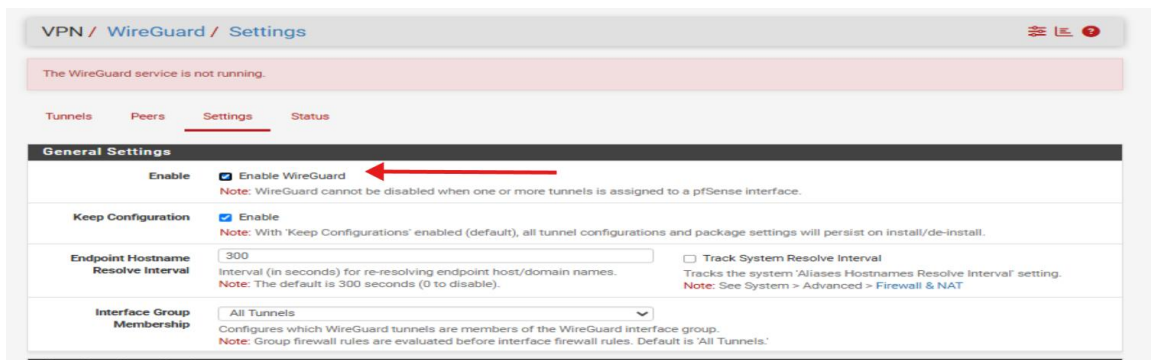


I create a tunnel for the VPN.

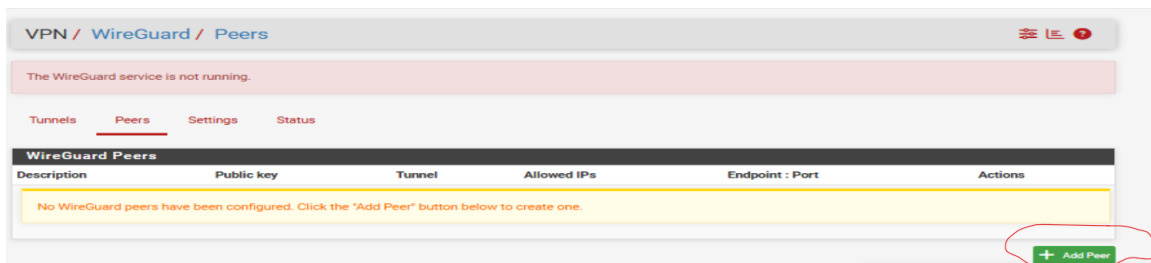


This is the VPN network.

SAVE Tunnel



I enable it → SAVE → APPLY CHANGES



I add a peer.

The screenshot shows the 'Peer Configuration' section of the pfSense interface. Red arrows point to the following fields:

- Enable Peer:** A checkbox that is checked. A note below it says: "Note: Uncheck this option to disable this peer without removing it from the list."
- Tunnel:** A dropdown menu set to 'tun_wg0 (WG_Server)'. A note below it says: "WireGuard tunnel for this peer. (Create a New Tunnel)"
- Description:** A text field containing 'Windows_Client'. A note below it says: "Peer description for administrative reference (not parsed)."
- Dynamic Endpoint:** A checkbox that is checked. A note below it says: "Note: Uncheck this option to assign an endpoint address and port for this peer."
- Keep Alive:** A text field containing '25'. A note below it says: "Interval (in seconds) for Keep Alive packets sent to this peer. Default is empty (disabled)."
- Public Key:** A text field containing a long alphanumeric string. A note below it says: "WireGuard public key for this peer."
- Pre-shared Key:** A text field with a 'Generate' button next to it. A note below it says: "Optional pre-shared key for this tunnel. (Copy)"
- Address Configuration:** A section with a 'Hint' and 'Allowed IPs' field. The 'Allowed IPs' field contains '10.100.100.2 / 32'. A note below it says: "IPv4 or IPv6 subnet or host reachable via this peer."

The IP was entered incorrectly here → I corrected it afterward.

This screenshot shows the same 'Peer Configuration' page, but with the 'Public Key' field updated to a new alphanumeric string. The 'Generate' button is still present next to the 'Pre-shared Key' field.

If not checked (Enabled), pfSense won't recognize that client and won't accept the VPN connection.

WG_Server → the peer will connect through this server.

VPN clients often get their IP via DHCP and their IP changes over time. With Dynamic Endpoint enabled, pfSense automatically accepts the new IP.

25 → this is WireGuard's recommended default keepalive value. It prevents the connection from dropping for devices behind NAT.

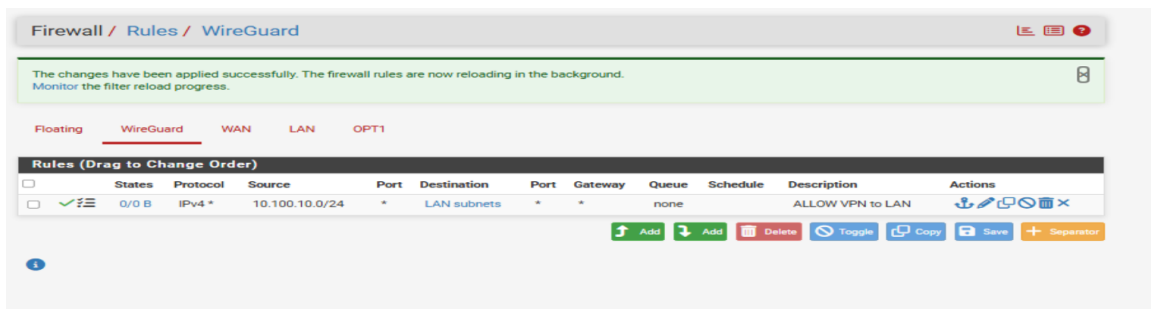
Windows Client IP → 10.100.100.2

SAVE → APPLY CHANGES

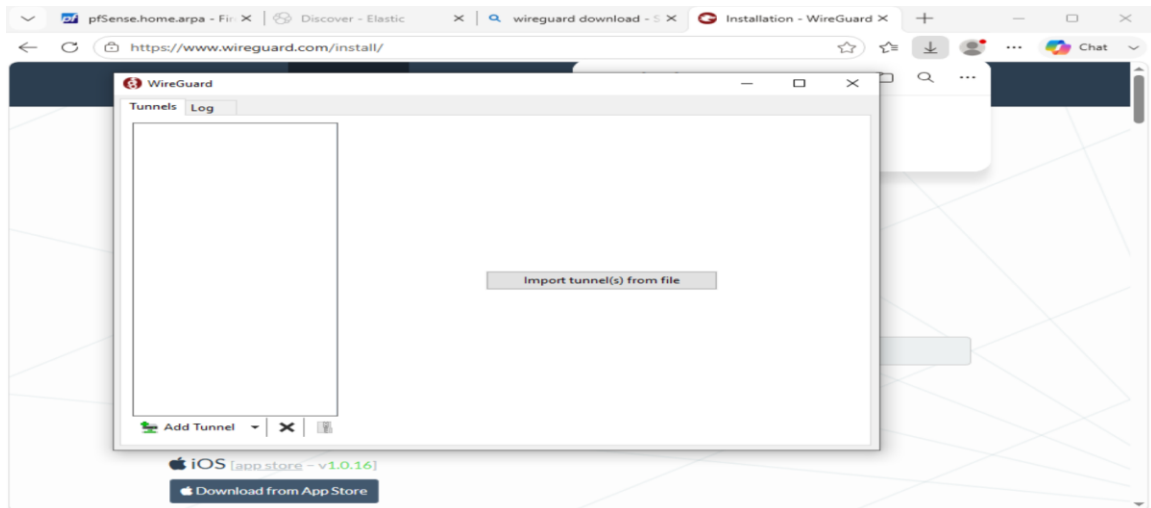
The screenshot shows the 'Firewall / Rules / WAN' page. A green message at the top states: "The changes have been applied successfully. The firewall rules are now reloading in the background. Monitor the filter reload progress." Below this, the 'Rules (Drag to Change Order)' table is visible. A new rule has been added at the bottom:

States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
0/15 KIB	*	RFC 1918 networks	*	*	*	*	*	*	Block private networks	[Settings]
0/0 B	*	Reserved Not assigned by IANA	*	*	*	*	*	*	Block bogon networks	[Settings]
0/0 B	IPv4 UDP	*	*	WAN address	51820	*	none		ALLOW WIREGUARD VPN	[Add] [Delete] [Toggle] [Copy] [Save] [Separator]

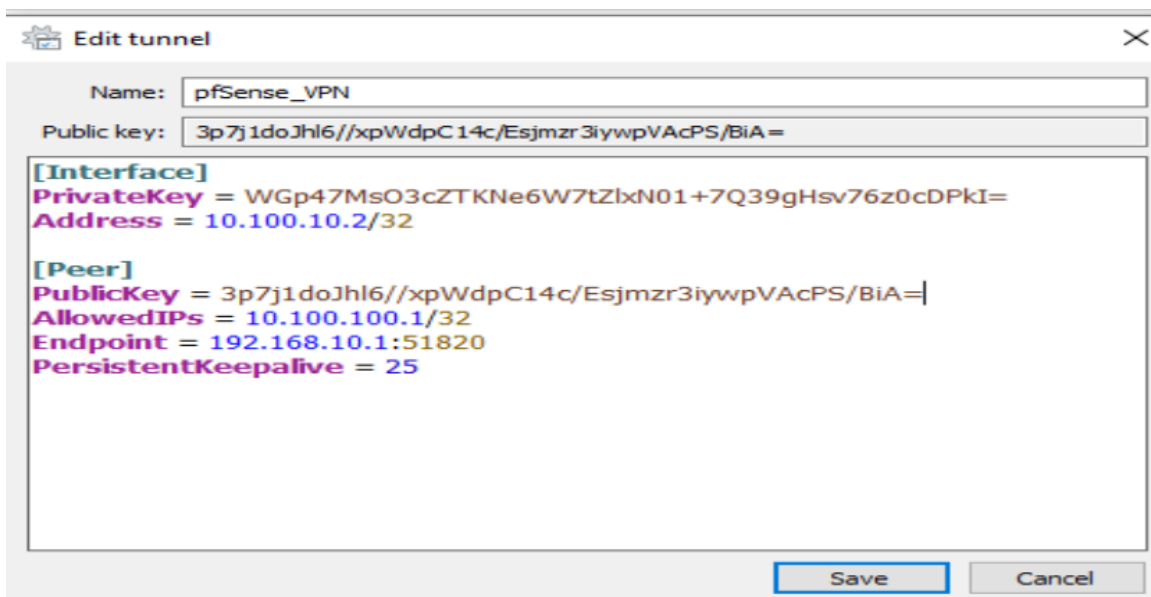
I add a rule → this rule allows UDP 51820 traffic from the VPN client to reach pfSense.



Now I install WireGuard on the Windows VM.



Add Tunnel → Add empty Tunnel



SAVE

